

# 07. Automatic Speech Recognition



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# Outline

7.1. Overview

7.2. System Architecture

7.3. ASR Techniques

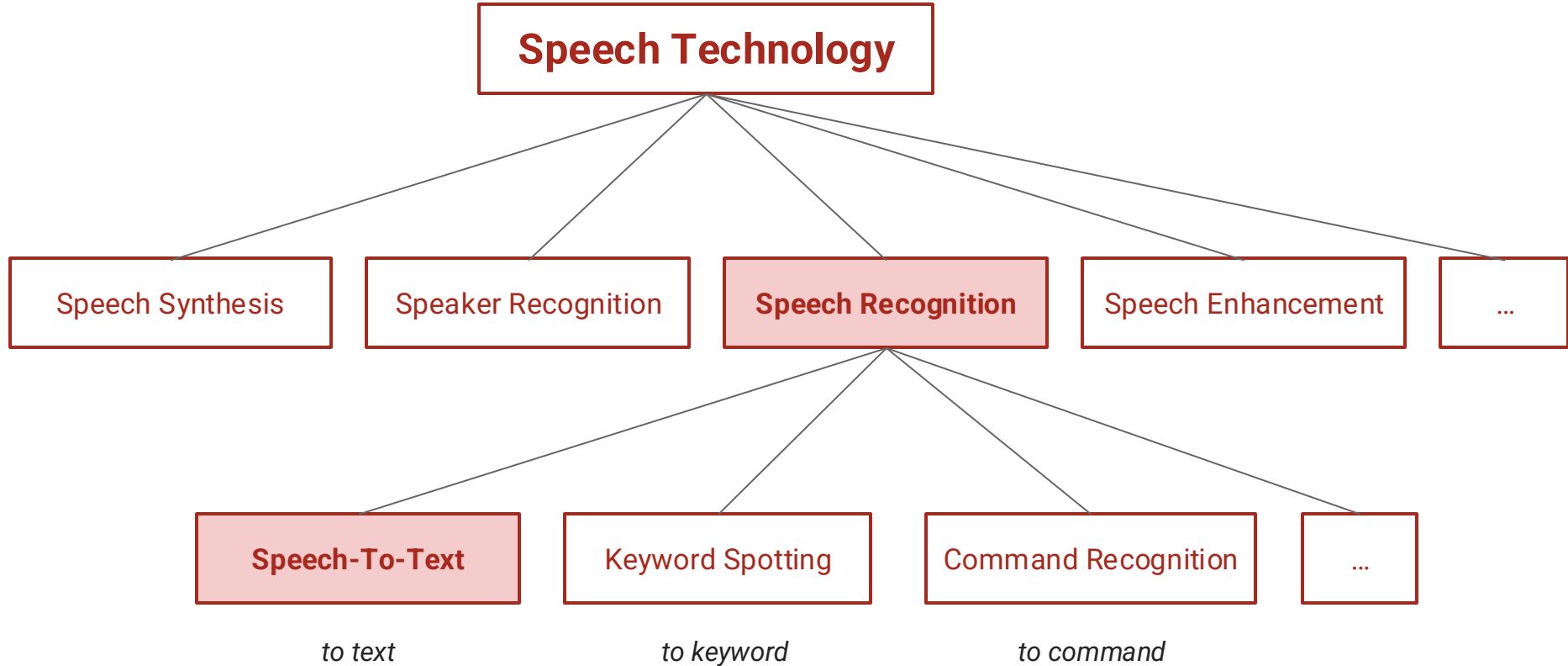
7.4. Challenges and Solutions

## 7.1 Overview

- ❖ Automatic Speech Recognition - Nhận dạng tiếng nói
  - Quá trình nhận diện và xử lý tiếng nói của con người thành những khía cạnh khác nhau
- ❖ Speech-To-Text (STT) - Chuyển từ tiếng nói thành văn bản
  - Một loại nhận dạng tiếng nói
  - Quá trình chuyển đổi từ đoạn âm thanh lời nói thành văn bản tương ứng



## 7.1. Overview



# History of Automatic Speech Recognition

- 
- 1950s–60s: **Template Matching**. Audrey, limited **digits**, single speaker
  - 1970s : **Isolated Word**. DARPA funding, small vocabulary, feature engineering
  - 1980s : **HMM**. Modular pipeline, continuous speech, statistical modeling
  - 1990s : **GMM-HMM**. Dictation tools, speaker independence, large vocabularies
  - 2000s : **Advanced ML**. Context modeling, beam search, robust lexicons
  - 2010s : **Deep Learning**. End-to-end models (CTC, RNN-T), DNN-based AMs
  - 2020s : **SSL + Transformers**. Pretrained models (Whisper, Wav2Vec 2.0), multilingual, open-source.

## ASR Application in Real-life?



<https://wall.sli.do/event/wPmQxU18pEZmnxHpv1Uo3m?section=c1a5f979-f9fd-41df-8ddd-d38a83ccd7e8>

# Applications of Speech Recognition



**Customer  
Service**



**Virtual  
Assistant**



**Transcription  
Services**



**Accessibility  
Features**



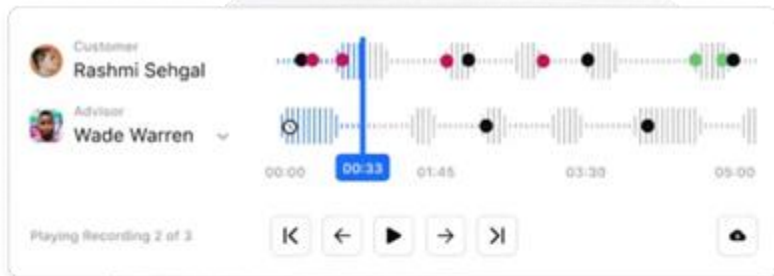
**Smart  
Homes**

# ASR Application in real-life

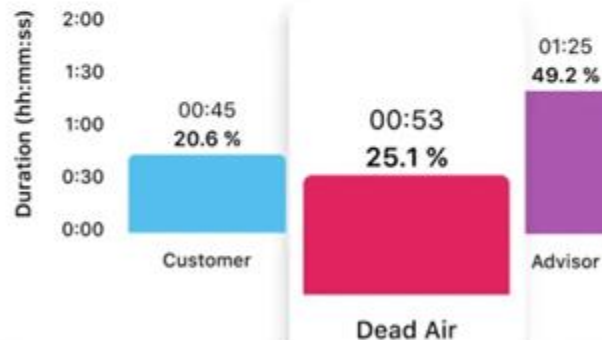
- ❖ Voice Assistants
- ❖ Call Center Automation
- ❖ Dictation / Voice Typing
- ❖ Subtitling
- ❖ Meeting Transcription
- ❖ Education
- ❖ In-car Voice Control
- ❖ Multilingual Interface



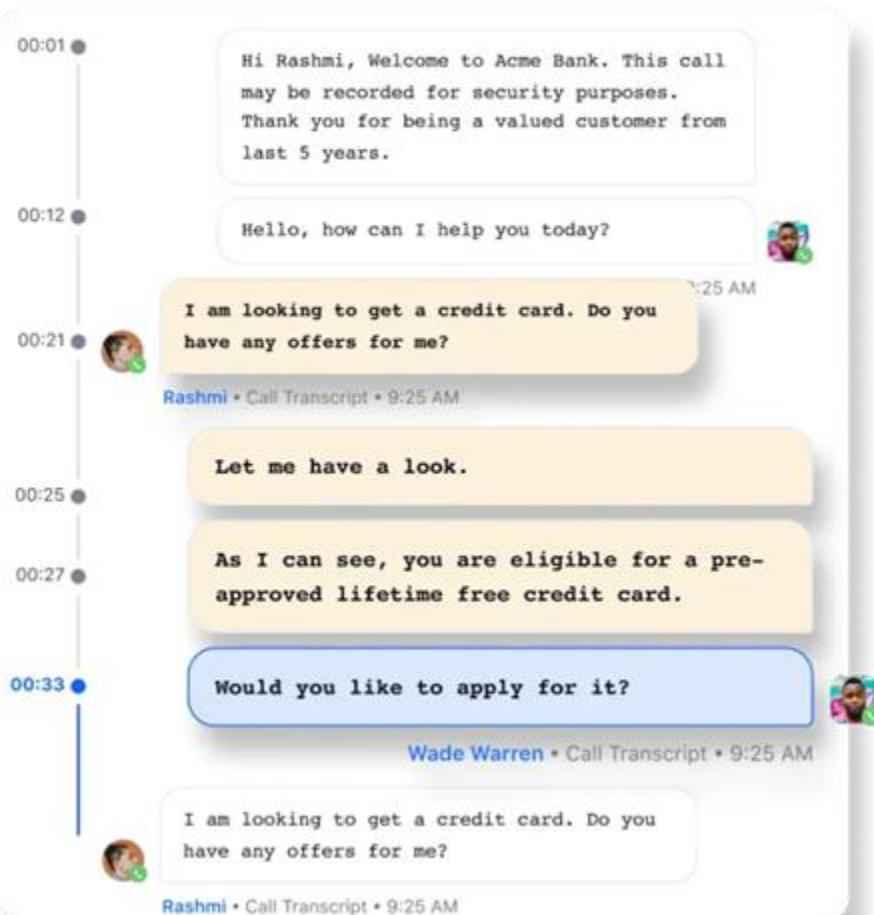




### Time Utilization

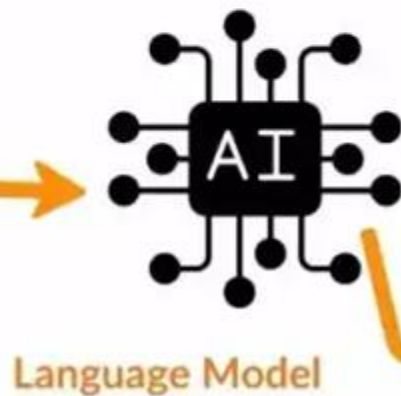
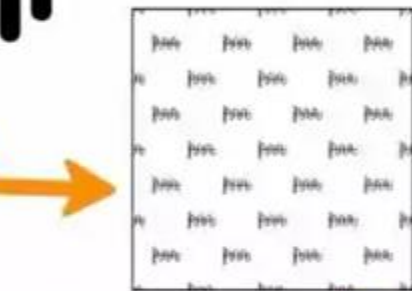


## Call Transcript



# Current State of ASR

| English Tasks                                         | WER% |
|-------------------------------------------------------|------|
| LibriSpeech audiobooks 960hour clean                  | 1.4  |
| LibriSpeech audiobooks 960hour other                  | 2.6  |
| Switchboard telephone conversations between strangers | 5.8  |
| CALLHOME telephone conversations between family       | 11.0 |
| Sociolinguistic interviews, CORAAL (AAL)              | 27.0 |
| CHiMe5 dinner parties with body-worn microphones      | 47.9 |
| CHiMe5 dinner parties with distant microphones        | 81.3 |
| Chinese (Mandarin) Tasks                              | CER% |
| AISHELL-1 Mandarin read speech corpus                 | 6.7  |
| HKUST Mandarin Chinese telephone conversations        | 23.5 |

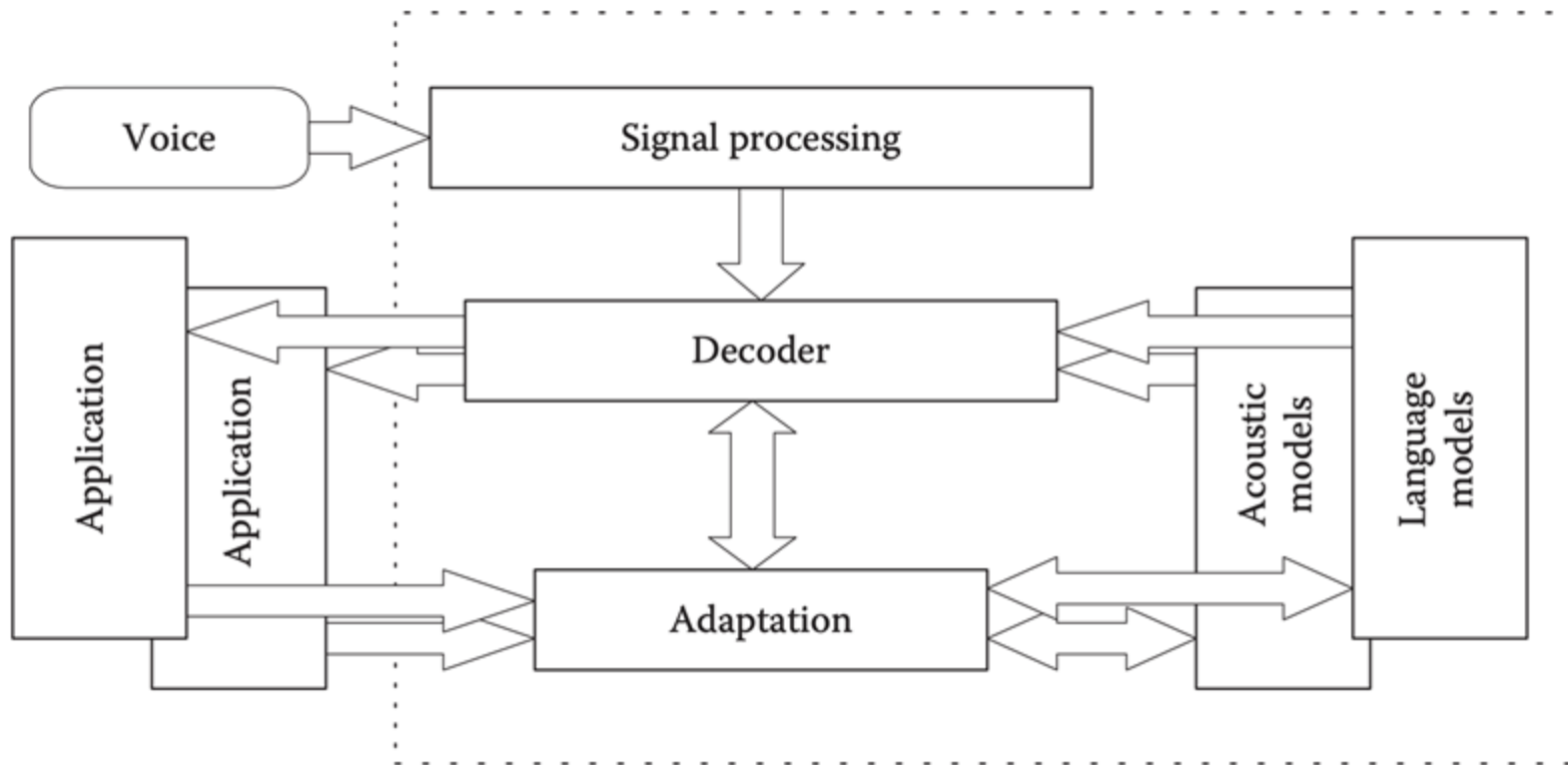


The Process of  
Speech Recognition

# ASR: Acoustic Model & Language Model



## 7.2 Architecture of ASR System

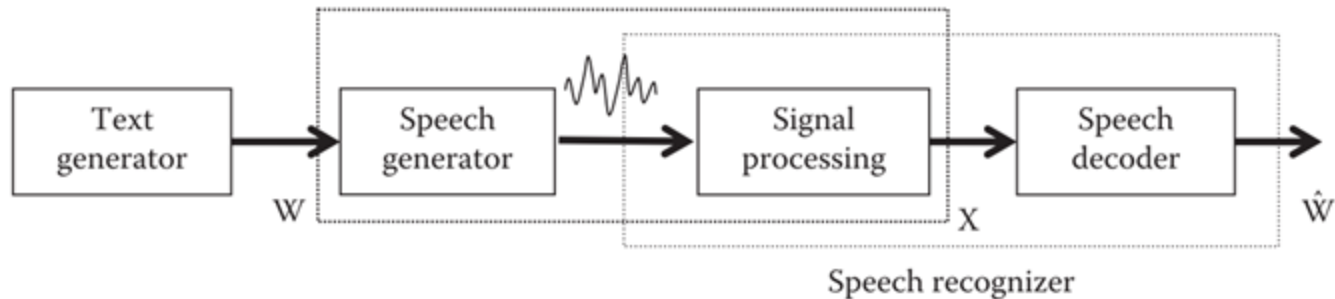


*Basic components of an ASR system.*

# Main Components

- **Acoustic Model:** Representation of knowledge about **acoustics**, **phonetics**, **microphone** and **environment** variability, **gender** and **dialect** differences among **speakers**, etc.
- **Language Model:** System's knowledge of what constitutes a **possible word**, what words are likely to **co-occur**, and in what **sequence**.
- **Decoder:** Generate the **word sequence** that has the maximum posterior probability for the input feature vectors extracted from speech signal.
- **Adaptation:** Modify either the acoustic or language models to **improve performance**

# ASR Problem



- **Input:** Acoustic observation or feature vector sequence  $\mathbf{X} = X_1 X_2 \dots X_n$
- **Output:** Corresponding word sequence  $\hat{\mathbf{W}} = w_1 w_2 \dots w_m$   
having maximum posterior probability  $P(W|X)$

$$\hat{\mathbf{W}} = \arg \max_{\mathbf{w}} P(\mathbf{W}|\mathbf{A}) = \arg \max_{\mathbf{w}} \frac{P(\mathbf{W})P(\mathbf{A}|\mathbf{W})}{P(\mathbf{A})} \stackrel{\text{Fixed } X}{=} \arg \max_{\mathbf{w}} P(\mathbf{W})P(\mathbf{X}|\mathbf{W})$$

- $P(W)$ : Language Model  
~ Lexical and grammatical complexity and variations of spoken language
- $P(X|W)$ : Acoustic Modeling ~ Phonetic Modeling  
~ Account speaker variations, pronunciation variations, environmental variations, and context-dependent phonetic coarticulation variations

## Best Word Sequence for Conventional Models

- $\lambda_1$ : Language model scaling factor (LMSF)
  - To recalibrate the trade-off between acoustics and common word sequences since the AM tends to underestimate  $P(X|W)$  due to the independence assumption on phone probabilities

$$\hat{W} = \operatorname{argmax}_W P_{AM}(X|W) P_{LM}(W)^{\lambda_1}$$

$$\hat{W} = \operatorname{argmax}_W \log P_{AM}(X|W) + \lambda_1 \log P_{LM}(W)$$



# Language Model

- Conventional models implicitly learn a language model for the output domain of characters from their training data
  - Training data (speech paired with text transcriptions) may not include sufficient text to train a good language model.
- Improve a model at least slightly by incorporating a very large language model

The roses in the vase by the door ?

Competing answers: is, are

$P(\text{The roses in the vase by the door are})$

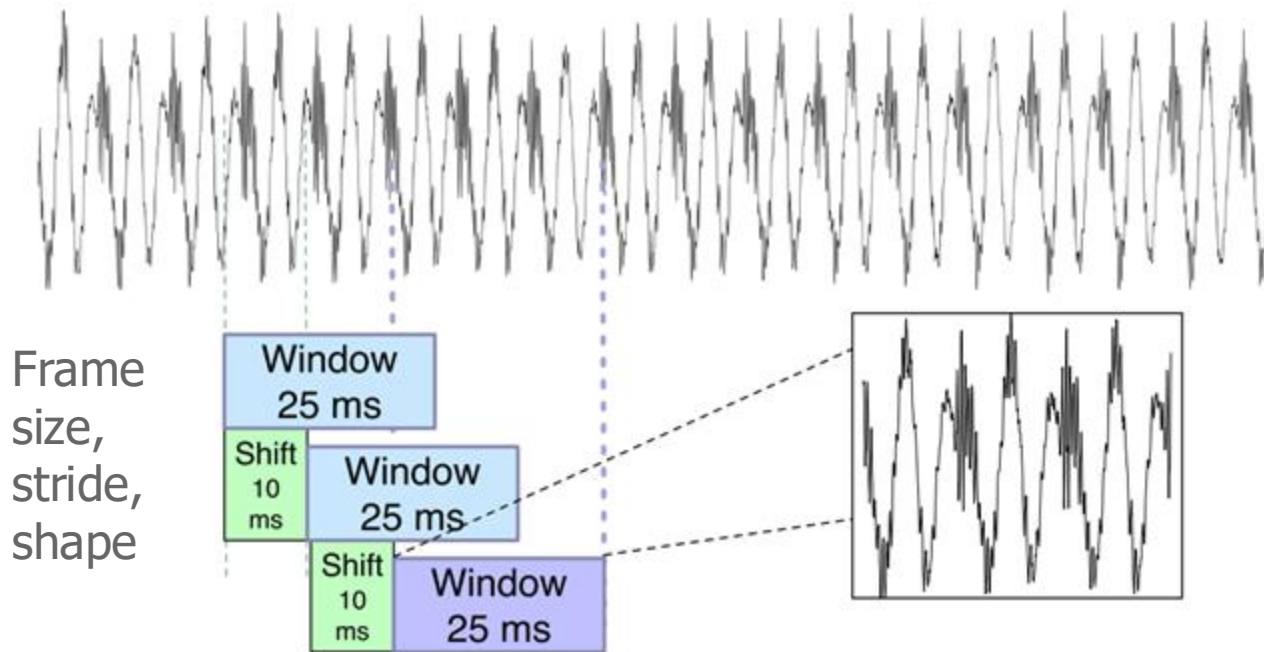
$P(\text{The roses in the vase by the door is})$

Is the correct answer  
ranked higher?

$P(\dots \text{are}) > P(\dots \text{is})?$

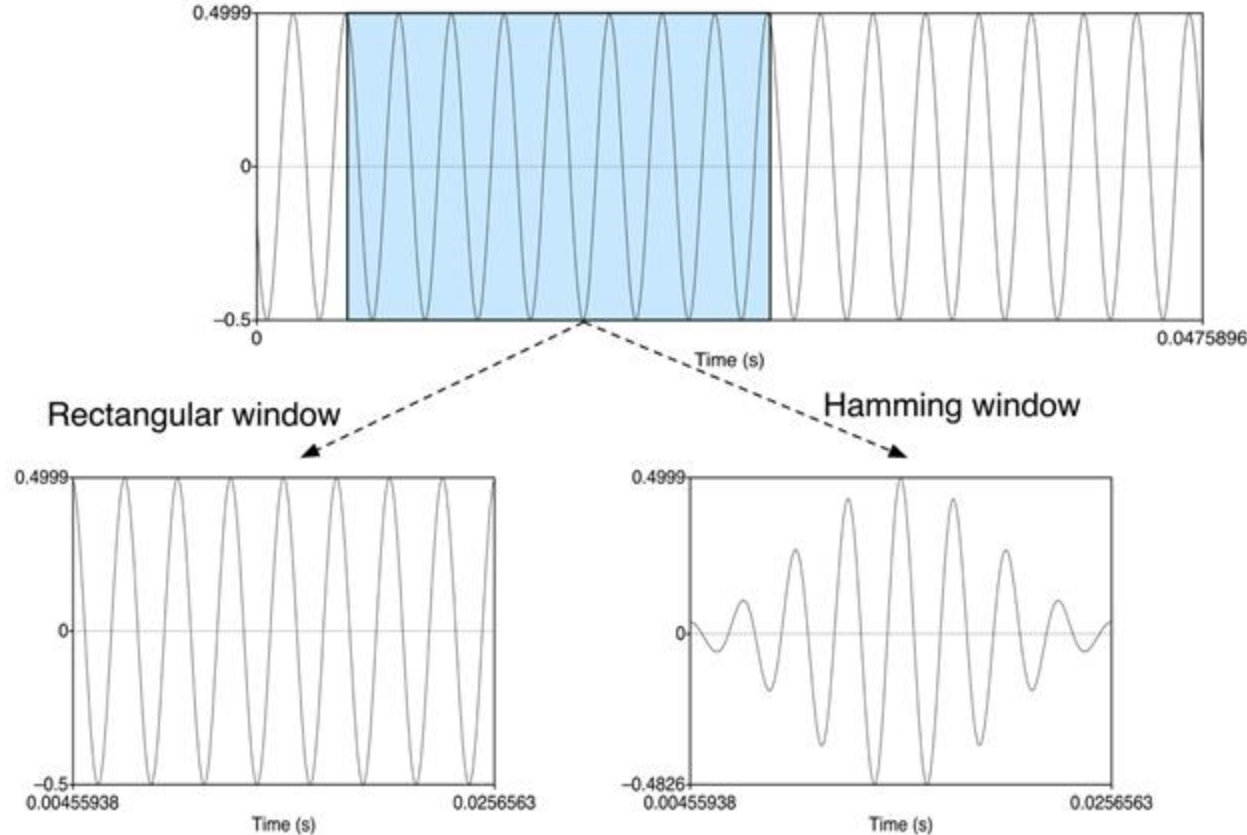
# Windowing to reduce spectral leakage

*Its statistical properties are constant within this region*  
→ *Extract acoustic feature vectors for each frame*



Windowing, showing a 25 ms rectangular window with a 10ms stride.

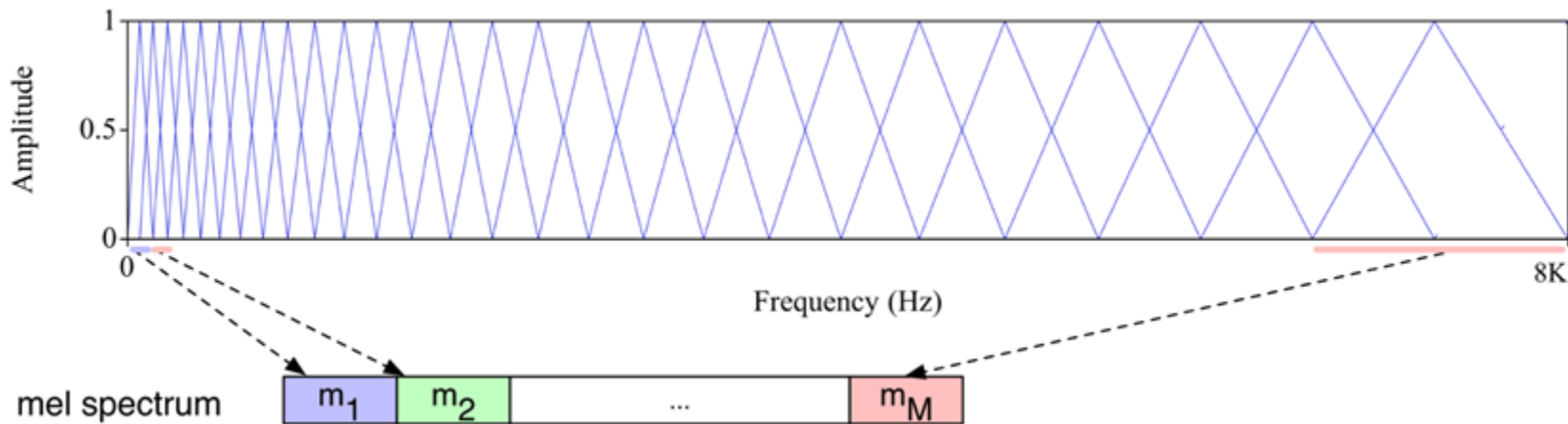
# Hamming window



Hamming window **shrinks** the values of the signal **toward zero** at the window **boundaries**, **avoiding discontinuities** compared to Rectangular

# Mel filter bank and Log

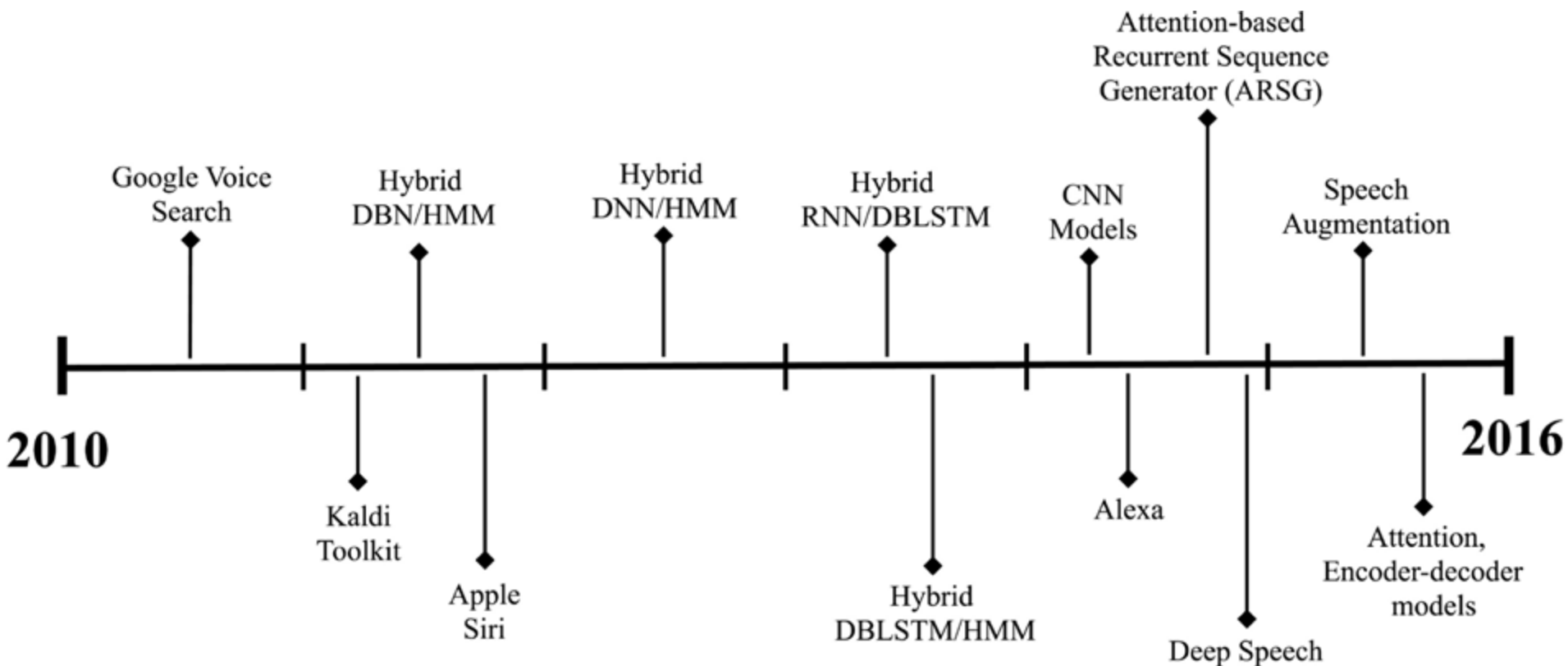
- **Mel scale:** Unit of pitch, auditory frequency scale for human hearing
- **Mel filter bank:** Bank of filters that collect energy from each frequency band, spread logarithmically: very fine resolution at low frequencies, and less resolution at high frequencies
  - Human hearing is not equally sensitive at all frequency bands, less sensitive at higher frequencies.



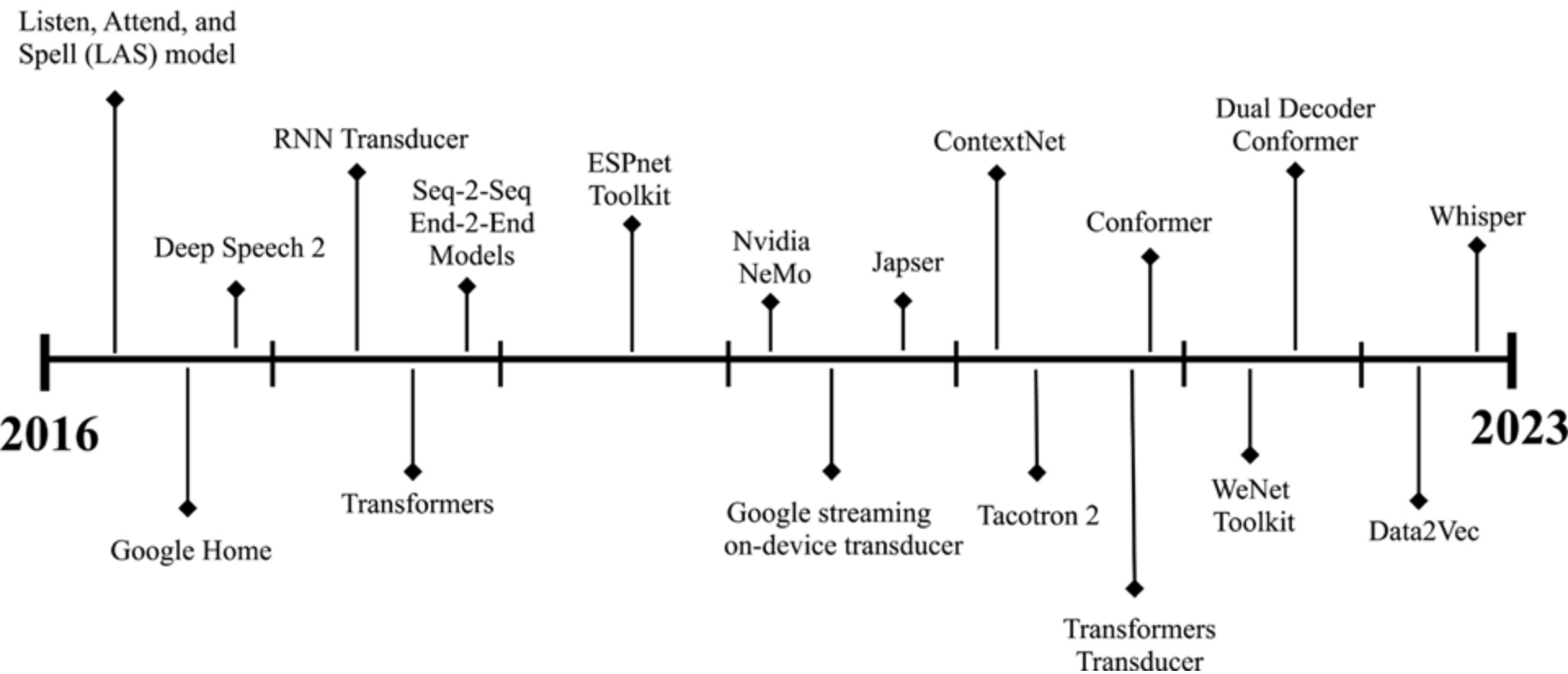
## 7.3. Speech Recognition Techniques

| Technique               | End-to-End | Streaming       | Popular Models/Tools                      |
|-------------------------|------------|-----------------|-------------------------------------------|
| HMM-based ASR           | ✗          | ✓               | Kaldi, HTK, CMU Sphinx, Julius            |
| CTC                     | ✓          | ✓ (with tricks) | DeepSpeech, Jasper, Kaldi (tools)         |
| RNN-T                   | ✓          | ✓               | Transducer-based models, TensorFlow ASR   |
| LAS (Attention-based)   | ✓          | ✗               | Listen, Attend, Spell                     |
| Transformer-based + SSL | ✓          | ✗/✓             | Wav2Vec 2.0, HuBERT, Whisper, SpeechBrain |

# From HMM to Attention Encoder-Decoder



# From LAS to SSL & Transformer



## Approaches / Techniques

1. Convolution Neural Network (CNN)

2. Recurrent Neural Network (RNN)

Long Short Term Memory (LSTM)

LSTM + RNN

3. Hybrid Architectures

CNN + RNN

LSTM + RNN + CNN

CNN + RNN + DNN

LSTM + CNN

DNN + RNN

ASR Techniques

4. Deep Neural Network (DNN)

5. Deep LSTM

11. Conformer

10. Transformer

Wav2Vec

Data2Vec

BERT

HuBERT

Whisper

9. Transfer Learning

8. Time Delayed Neural Network (TDNN)

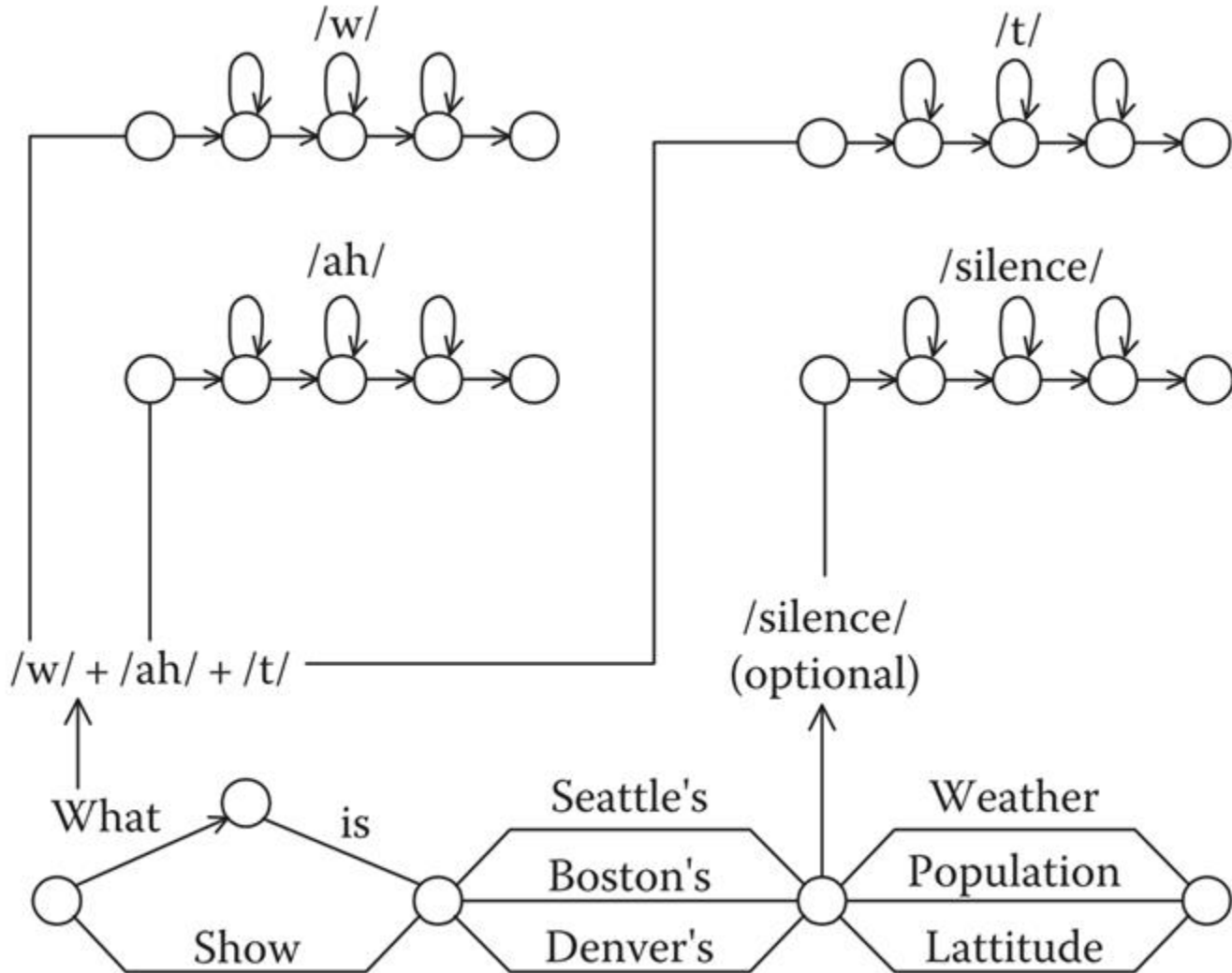
7. Deep Convolution Neural Network (DCNN)

6. Deep Belief Network (DBN)



## 7.3.1. HMM-based ASR

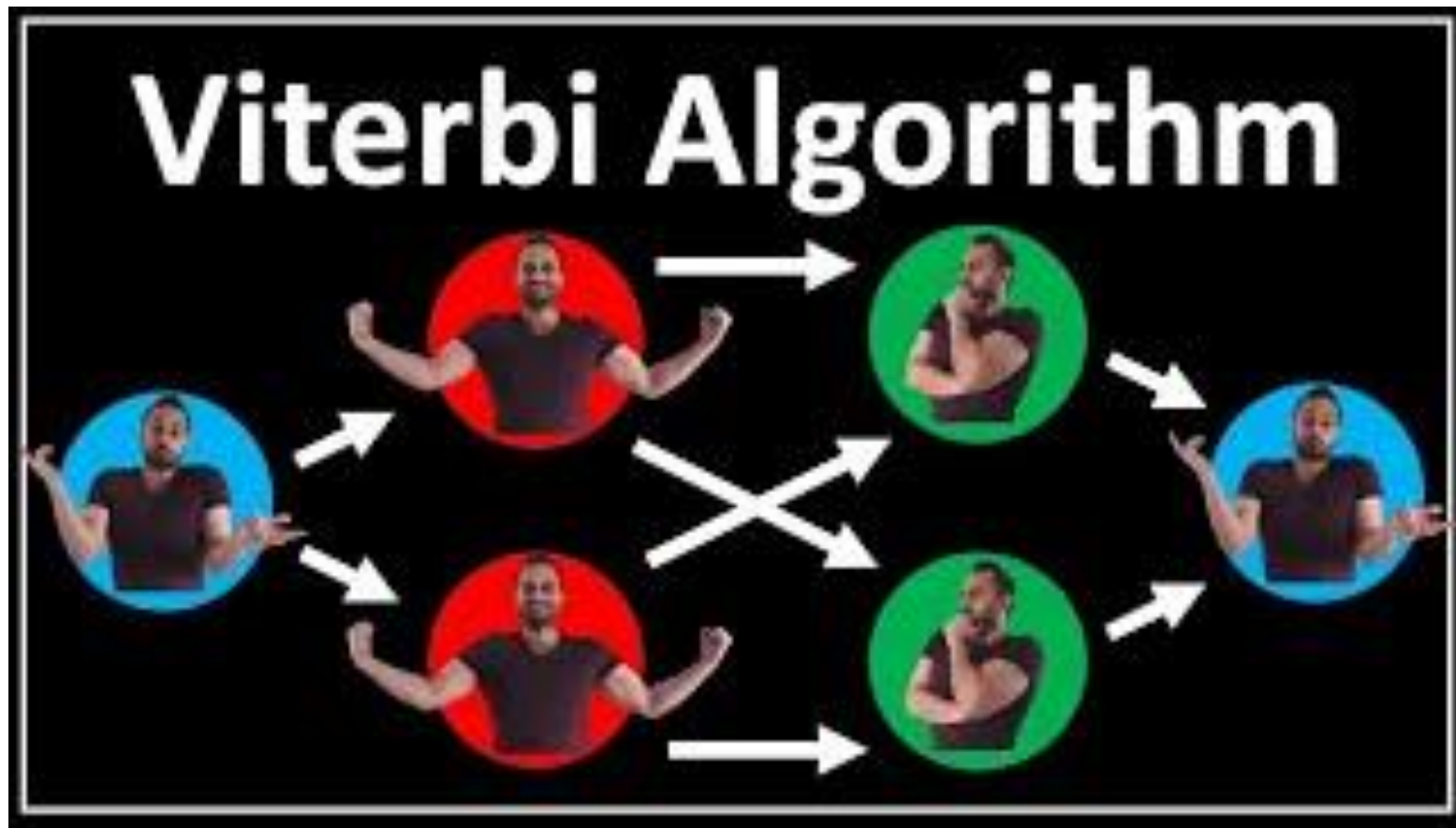
An illustration of how to compile a speech-recognition task with finite grammar into a composite HMM.



# Hidden Markov Model

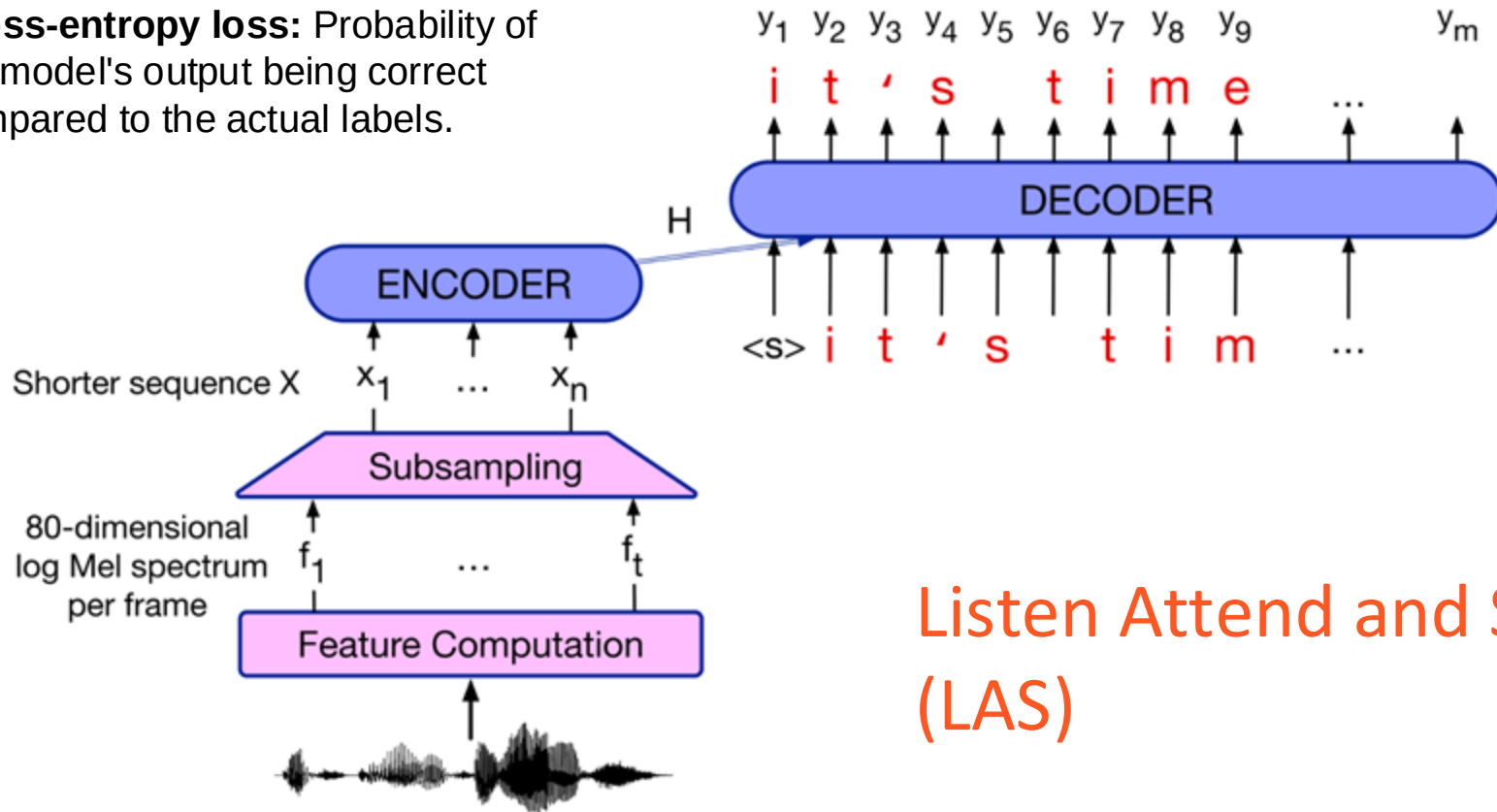


## Viterbi algorithm



## 7.3.2. Attention-based Encoder-Decoder ASR

**Cross-entropy loss:** Probability of the model's output being correct compared to the actual labels.



Listen Attend and Spell  
(LAS)

# Two particular issues of ASR

- **Length differences**

- Very long acoustic feature sequences (200 acoustic 10ms-frames) mapping to much shorter sequences of letters (5 letters)

→ **AED** has a special **compression** stage that shortens the acoustic feature sequence before the encoder stage as well as an **attention** mechanism focusing on different parts of the input while predicting the output.

- **Input-output mapping:**

- Hard to know exactly which part of X maps to which part of Y

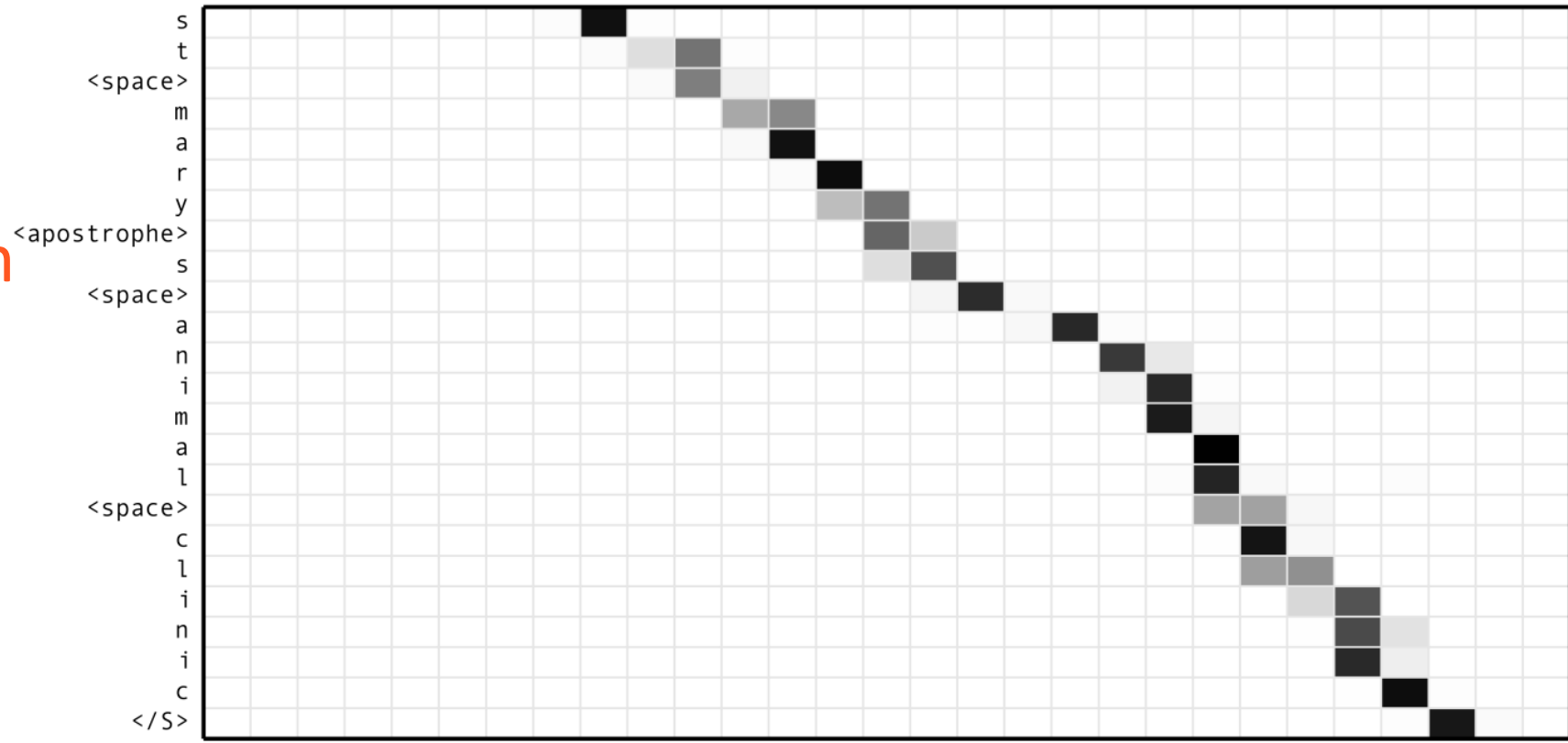
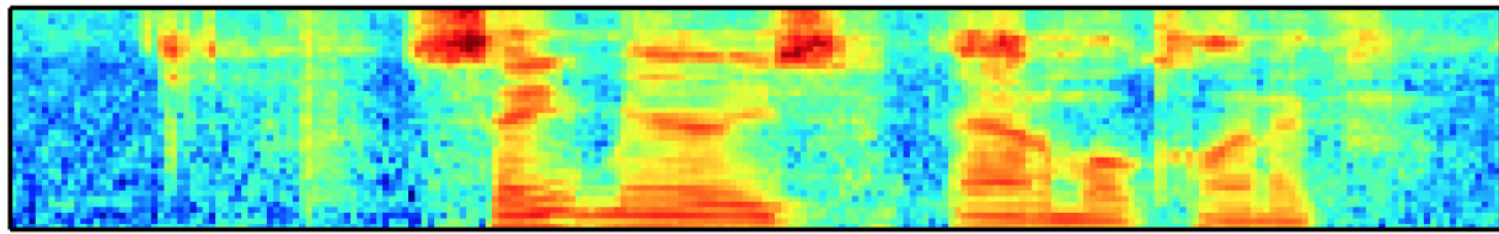
→ **AED** has an **attention** mechanism to align parts of the input with corresponding parts of the output, without requiring predefined structures.

## Problems of AED in ASR

- The attention operation is expensive for long input sequences
  - The complexity of attending to the entire input for every output is  $O(mn)$ , and for audio,  $m$  and  $n$  are big.
- Attention models cannot be run *online* (in real time), since the entire input sequence needs to be available before the decoder can attend
- Attention models also don't take advantage of the fact that, for speech recognition, the alignment between inputs and outputs is **monotonic**
  - i.e. if word A comes after word B in the transcript, word A must come after word B in the audio signal.
  - The fact that attention models lack this inductive bias seems to make them harder to train for speech recognition; it's common to add auxiliary loss terms to stabilize training.

Monotonic Alignment

Audio

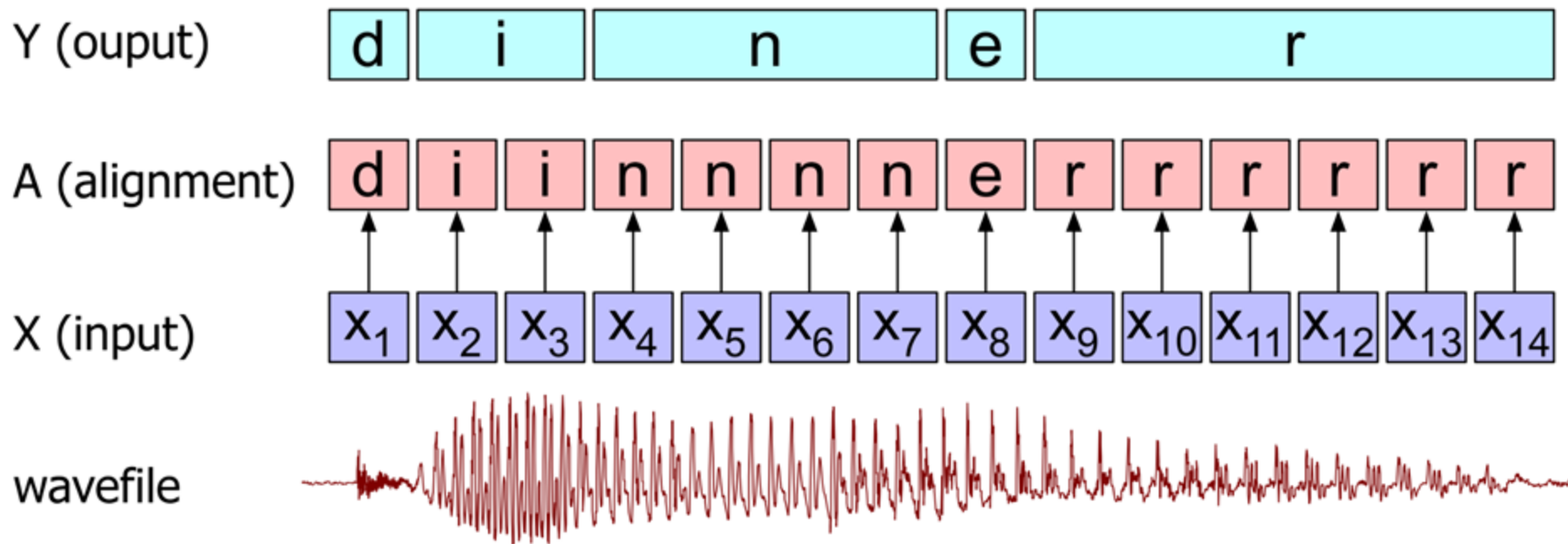


### 7.3.3. Connectionist Temporal Classification (CTC)

- Predict character sequences **without explicitly matching the input's length to the output**, enabling it to handle variable-length outputs and learn rare or uncommon words.
- **Algorithm:**
  - Output a single character for every frame of the input, so that the output is the same length as the input
  - Apply a collapsing function combining sequences of identical letters, resulting in a shorter sequence.
- **Loss function:**
  - Probability of all possible outputs to the actual outputs, including using a special symbol to represent gaps between words or parts of a word
  - Without requiring exact alignment between input and output sequences.

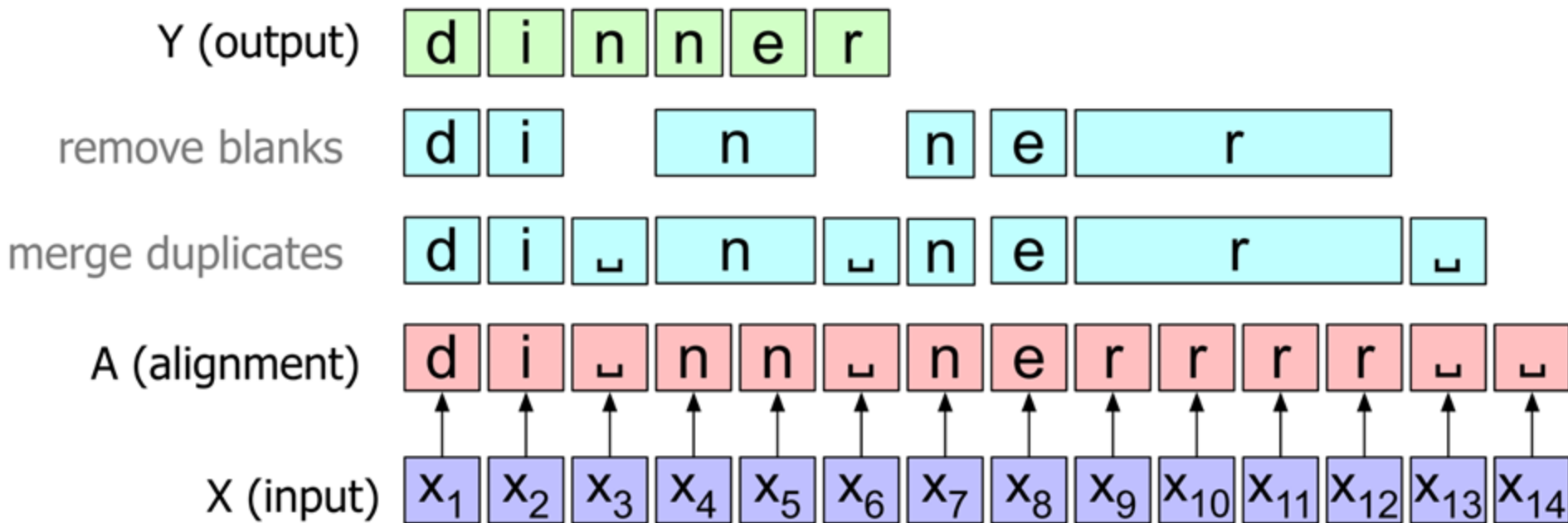


## Naive collapsing an alignment



*Naive algorithm for collapsing an alignment between input and letter*

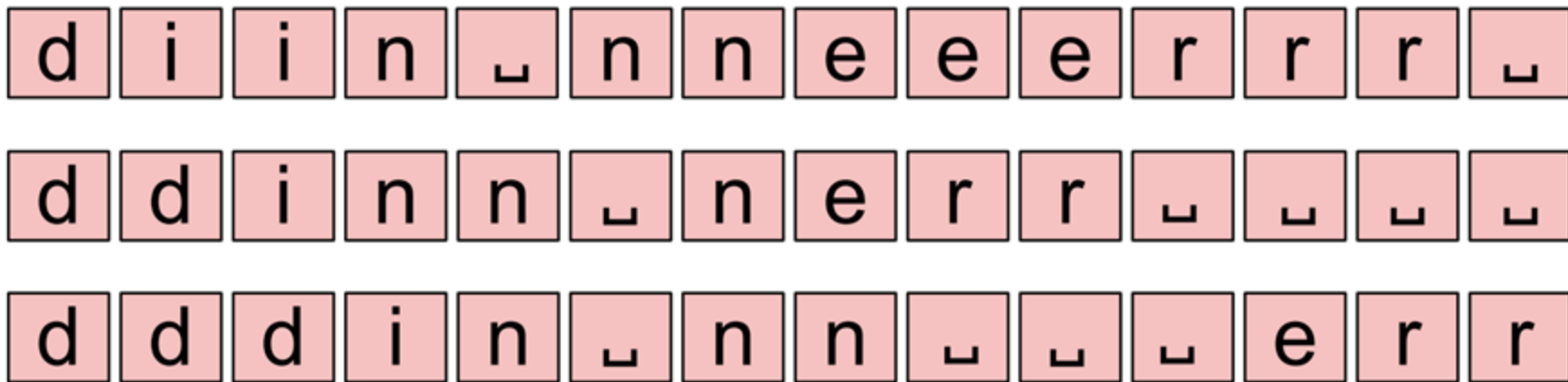
## Blank token in CTC alignment



*CTC adds to the transcription alphabet a special symbol for a blank*

# CTC Alignment

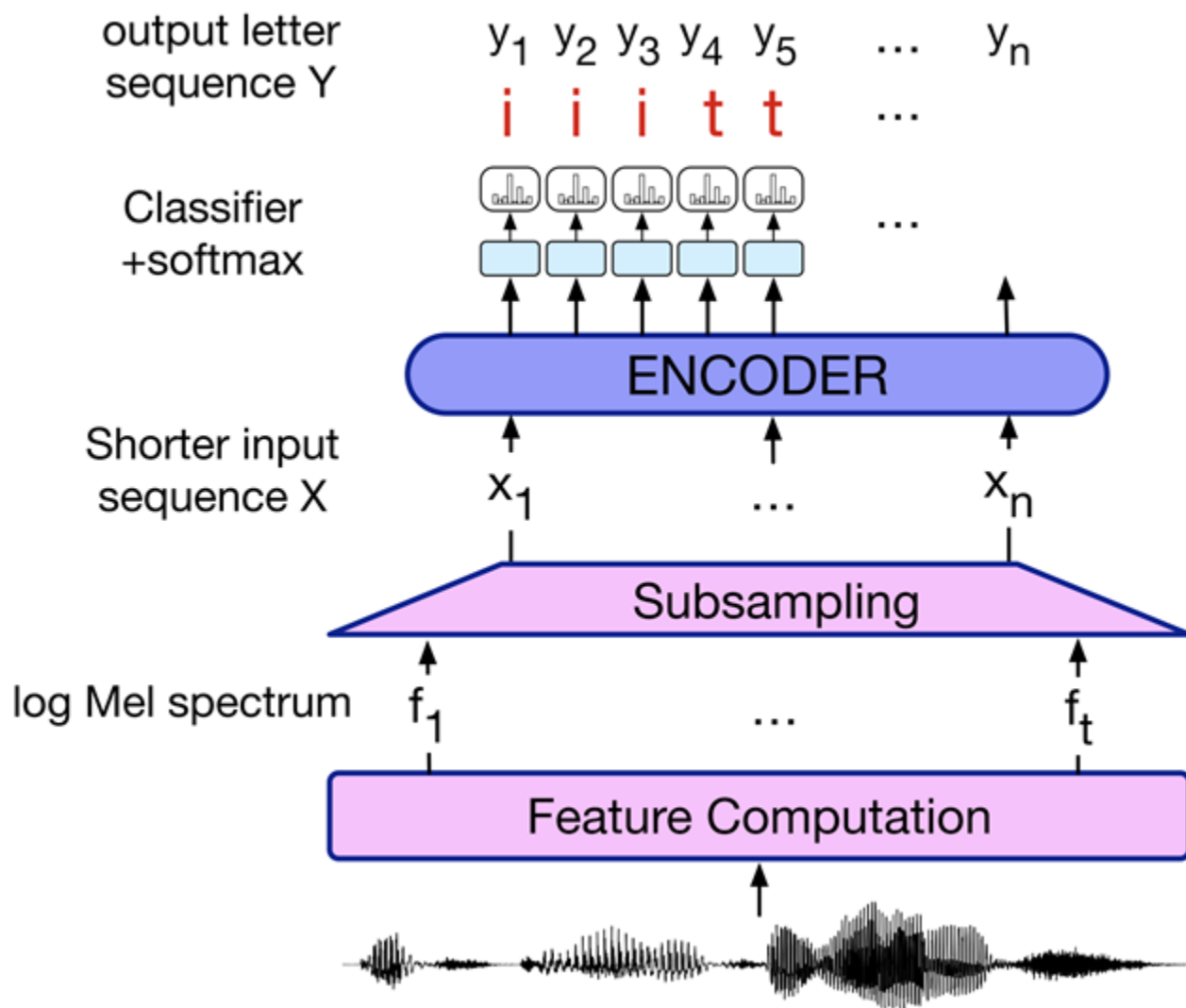
E.g. Different alignments that would produce the same output



## CTC Inference

~ Sequence-modeling

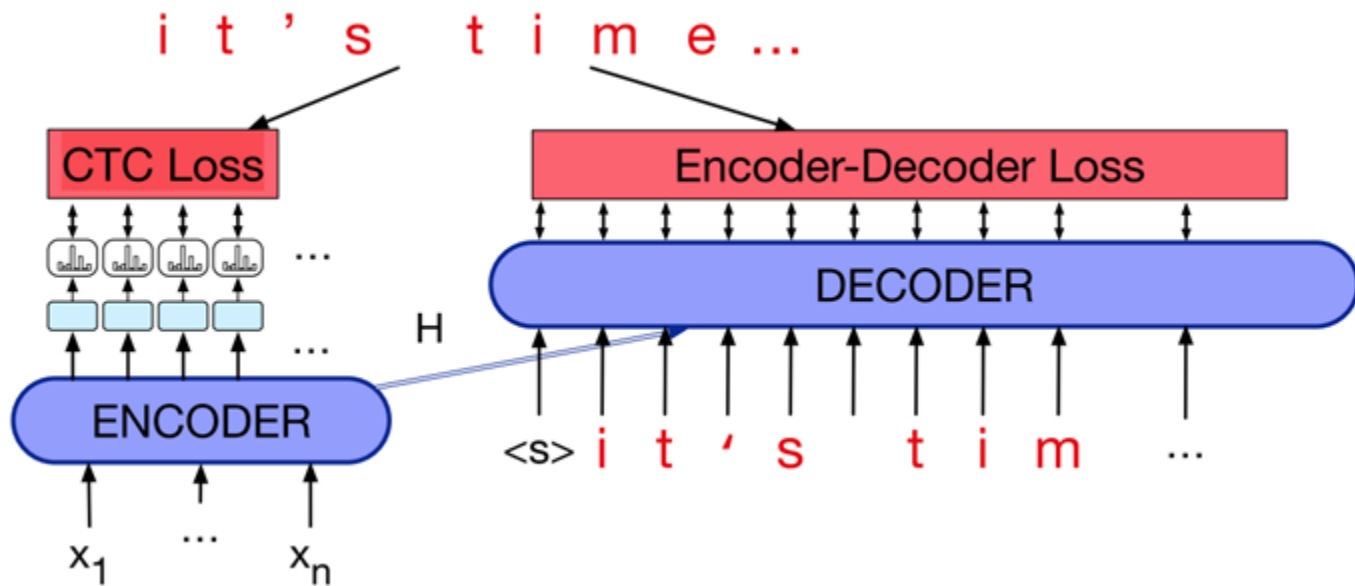
~ Encoder-only model,  
with decoding done by  
simple softmaxes over  
the character  
vocabulary at each  
time step



## Combining CTC and Encoder-Decoder

Combining cross-entropy loss from the encoder-decoder architecture, and the CTC loss

$$L = -\lambda \log P_{encdec}(Y|X) - (1 - \lambda) \log P_{ctc}(Y|X)$$

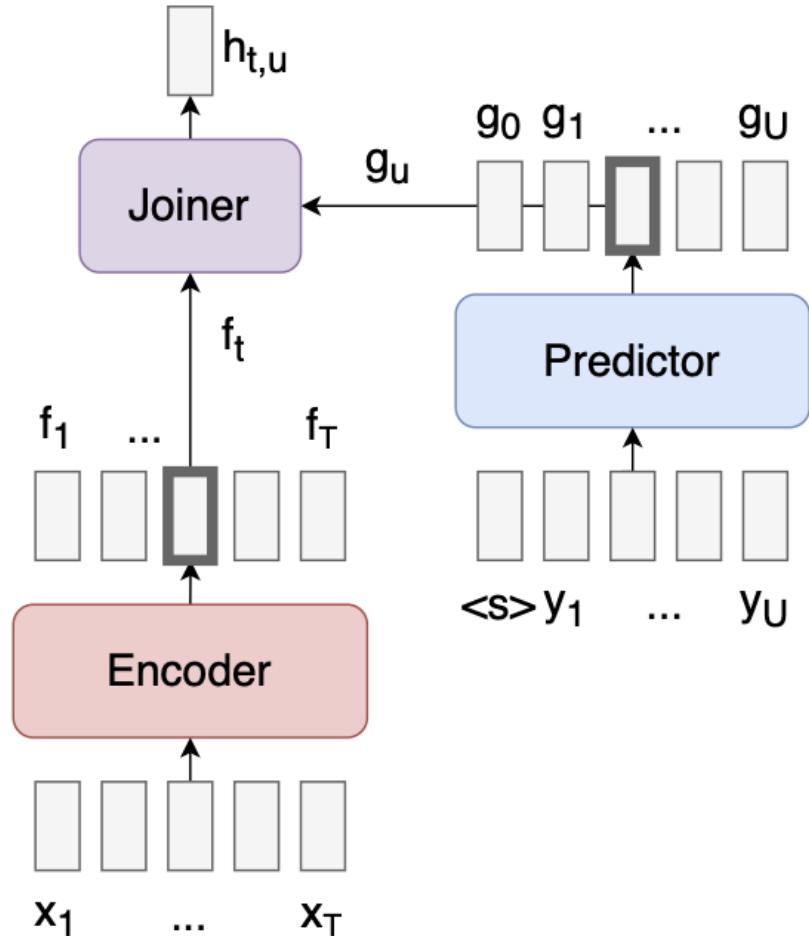


# Problems of CTC

- *Problem 1: The output sequence length has to be smaller than the input sequence length*
  - This might not seem like a problem for speech recognition, where length of output  $T$  is much larger than length of input  $U$ —but it prevents us from using a model architecture that does a lot of pooling, which can make the model a lot faster.
- *Problem 2: The outputs are assumed to be independent of each other*
  - The result is that CTC models often produce outputs that are obviously wrong, like “I eight food” instead of “I ate food”. Getting good results with CTC usually requires a search algorithm that incorporates a secondary language model.

# RNN-T

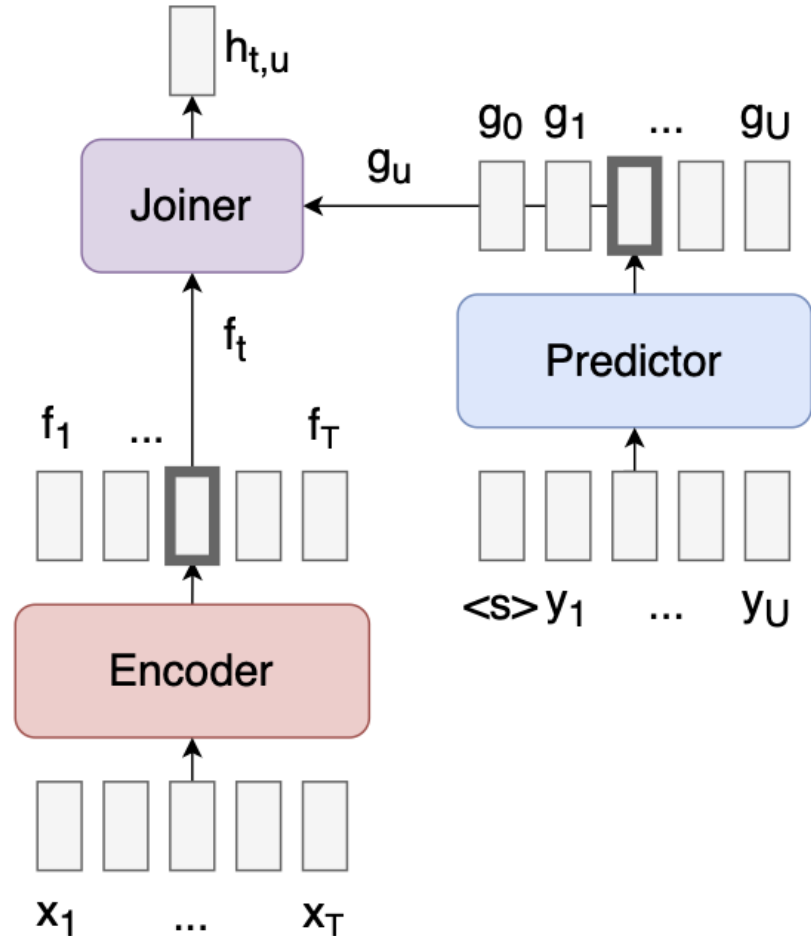
- RNN-T is an **end-to-end sequence-to-sequence model designed for ASR**. It **jointly learns the alignment between the input audio and the output text transcription** without requiring explicit frame-level alignment
  - Overcomes limitations of CTC which waits for a larger chunk or the full sentence, then collapses the output — introducing delay.
  - Allows for more flexible alignment between speech frames and text tokens
  - Supports streaming ASR, where transcriptions can be generated as the audio is being processed.



$$P_{AM}(X|W) = \frac{P_{RNNT}(W|X)P(X)}{P_{ILM}(W)}$$

### 7.3.4. Transducer model (RNN-T)

- The Transducer elegantly solves both problems associated with CTC, while retaining some of its advantages over attention models
  - It solves **Problem 1** by allowing multiple outputs for each input
  - It solves **Problem 2** by adding a predictor network and joiner network.

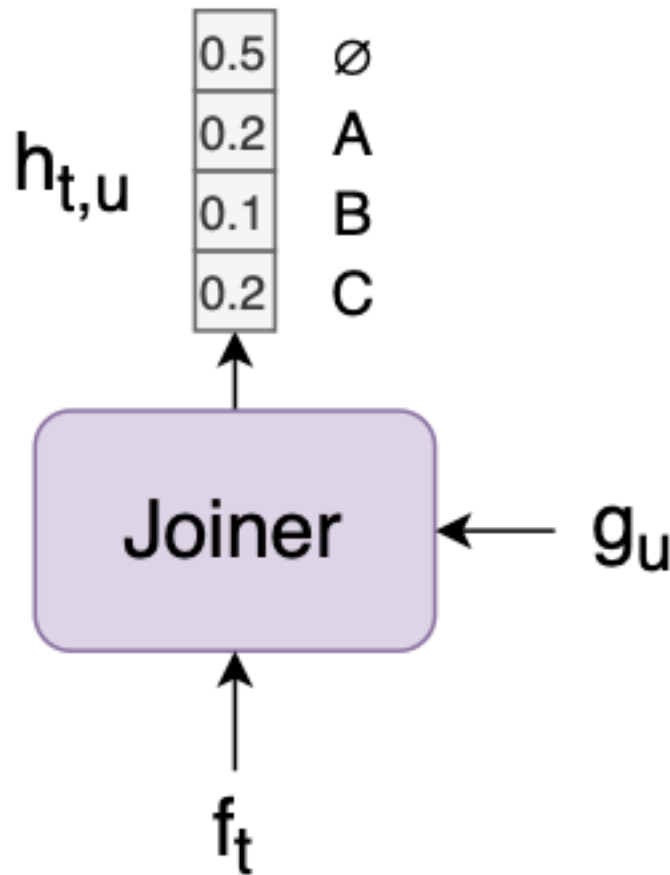


$$P_{AM}(X|W) = \frac{P_{RNNT}(W|X)P(X)}{P_{ILM}(W)}$$

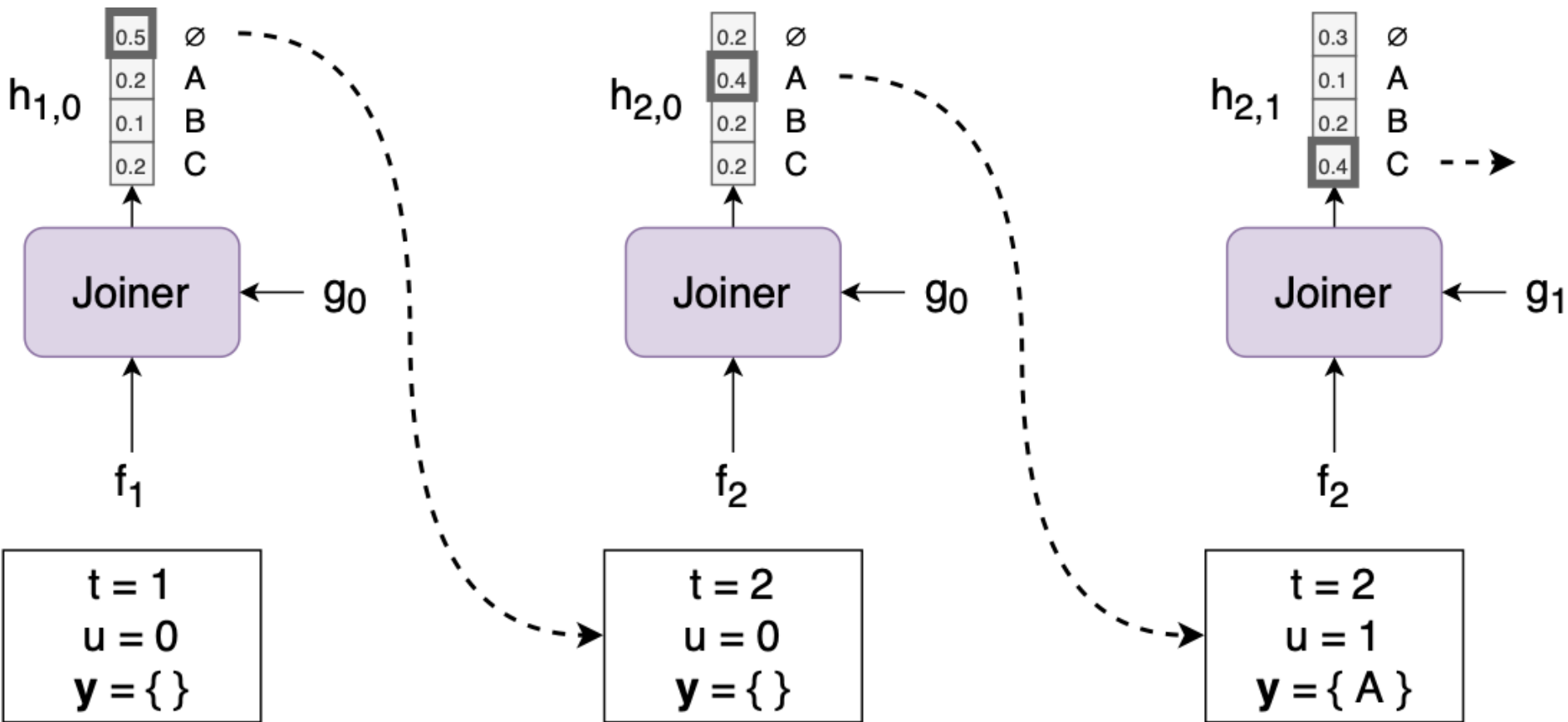


## RNN-T

- The predictor is autoregressive
  - it takes as input the previous outputs and produces features that can be used for predicting the next output, like a standard language model.
- The joiner is a simple feedforward network that combines the encoder vector  $f_t$  and predictor vector  $g_u$  and outputs a softmax  $h_{t,u}$  over all the labels, as well as a “null” output  $\emptyset$



## Simple greedy search algorithm



# Why RNN-T Is Suitable for Streaming ASR?

- Streaming-friendly
  - Processes audio frame-by-frame without needing the full sequence ahead of time
- No need for forced alignment
  - Learns alignment automatically (like CTC), no frame-level labels needed
- Supports incremental output
  - Can generate words while audio is still coming in — critical for low-latency ASR
- Monotonic decoding
  - Naturally maps well to the left-to-right nature of spoken language

# Why RNN-T Is Suitable for Streaming ASR?

- Token-by-token prediction
  - RNN-T can emit tokens during audio input, frame-by-frame.
  - CTC often needs to wait until the full utterance or chunk is processed before collapsing repeated tokens and blanks.
- Context-aware decoding
  - RNN-T uses a Prediction Network (decoder) that retains previous outputs, giving it language modeling capabilities during decoding.
  - CTC has no such decoder, so it lacks contextual understanding during prediction.
- Joint network enables streaming
  - The Joint Network combines outputs from the encoder (acoustic info) and prediction network (linguistic context) to decide what to emit — ideal for real-time decoding.

# CTC

vs.

# RNN-T

✗ Not ideal for real-time streaming

Assumes monotonic alignment, no decoder context

✗ No decoder, output depends only on encoder

Outputs after processing entire input or a large chunk

High latency (especially on long utterances)

Offline transcription (e.g., post-processing voice recordings)

✓ Specifically designed for streaming

Learns alignment using joint acoustic + linguistic context

✓ Has a **prediction network** that models output history

Can emit tokens **incrementally** as audio frames arrive

Low latency; suitable for real-time interaction

Real-time ASR (e.g., voice assistants, live captioning)

## Problems of RNN-T

- Latency
- Accuracy with long context
- Expensive training

## How to improve RNN-T?

- Replacing RNNs with More Efficient Architectures (Transformer/Conformer)
- Replacing the Prediction Network (Decoder) with lightweight Transformers
- Enhancing the Loss Function with CTC Loss, LM Loss, Alignment-aware Loss...
- Integrating External Language Models (LM)
- Multilingual and Code-Switching ASR

## External LM vs Internal LM (ILM) to RNN-T

- RNNT directly models the posterior  $P(W|X)$  directly
  - Unlike the AM in conventional systems which models the likelihood  $P(X|W)$
- $\lambda_1, \lambda_2$ : Language model scaling factor (LMSF)
  - To recalibrate the trade-off between acoustics and common word sequences since the AM tends to underestimate  $P(X|W)$  due to the independence assumption on phone probabilities

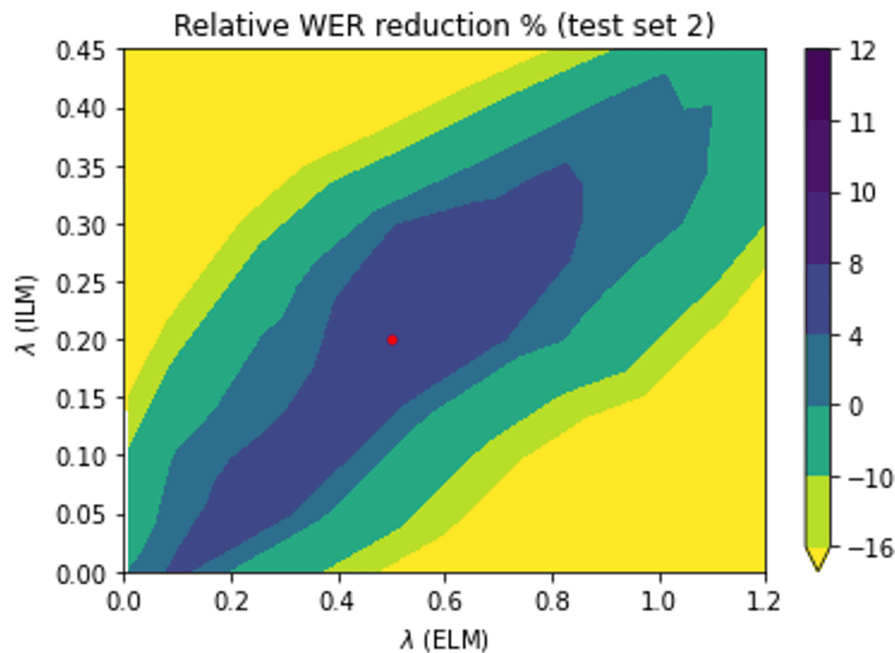
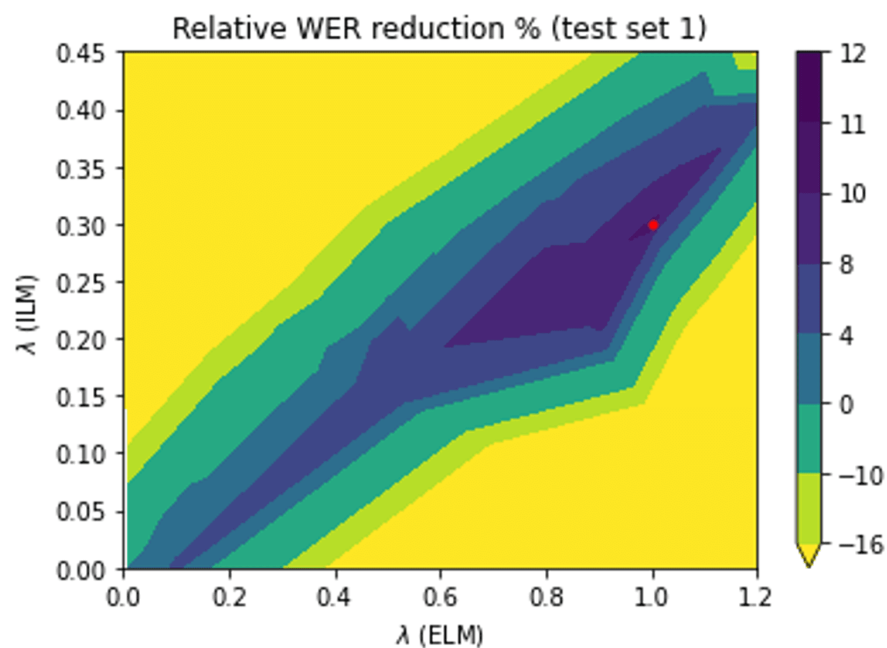
$$P_{AM}(X|W) = \frac{P_{RNNT}(W|X)P(X)}{P_{ILM}(W)} \quad \hat{W} = \operatorname{argmax}_W \frac{P_{RNNT}(W|X)}{P_{ILM}(W)^{\lambda_2}} P_{LM}(W)^{\lambda_1}$$

$$\hat{W} = \operatorname{argmax}_W \log P_{RNNT}(W|X) - \lambda_2 \log P_{ILM}(W) + \lambda_1 \log P_{LM}(W)$$



# Tuning LMSF

- More ILM-subtraction introduces the need for more external LM addition to take up the slack



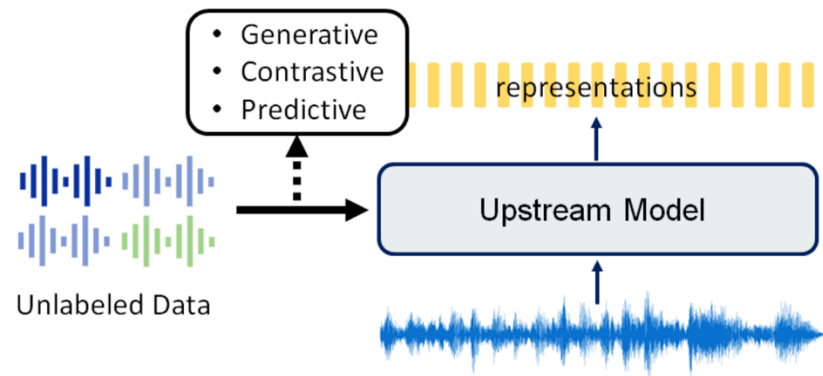
## 7.3.5. SSL + Transformer

- Self-Supervised Learning (SSL):
  - Pre-trains models on large amounts of **unlabeled data** to learn useful representations of speech.
  - Training the model to **predict masked parts** of the input or solve other self-defined tasks **without human annotations**
  - Pre-trained representations capture **general acoustic and phonetic** information
- Transformer
  - **Self-attention** mechanism to model relationships between different parts of a sequence
  - Can capture **long-range dependencies** in speech more effectively than traditional RNNs
  - Allows for **parallel processing** of the input sequence, leading to faster training times

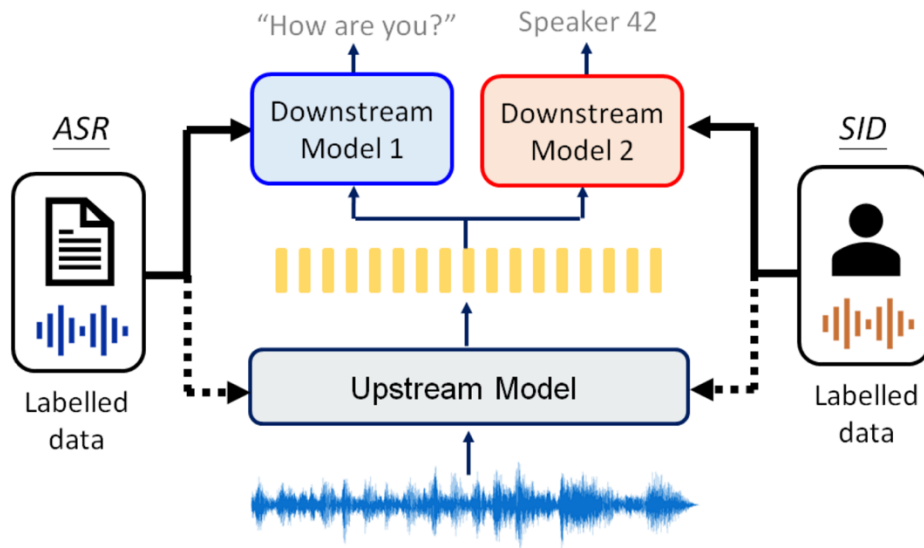
# Why Combine SSL with Transformers in ASR?

- Learn from unlabeled audio
  - SSL enables pretraining on large-scale untranscribed speech
- Improved generalization
  - Robust across speakers, accents, and domains
- Low-resource friendly
  - Performs well even with limited fine-tuning data
- Multilingual & zero-shot
  - Especially effective in models like Whisper or HuBERT

## Phase 1: Pre-train

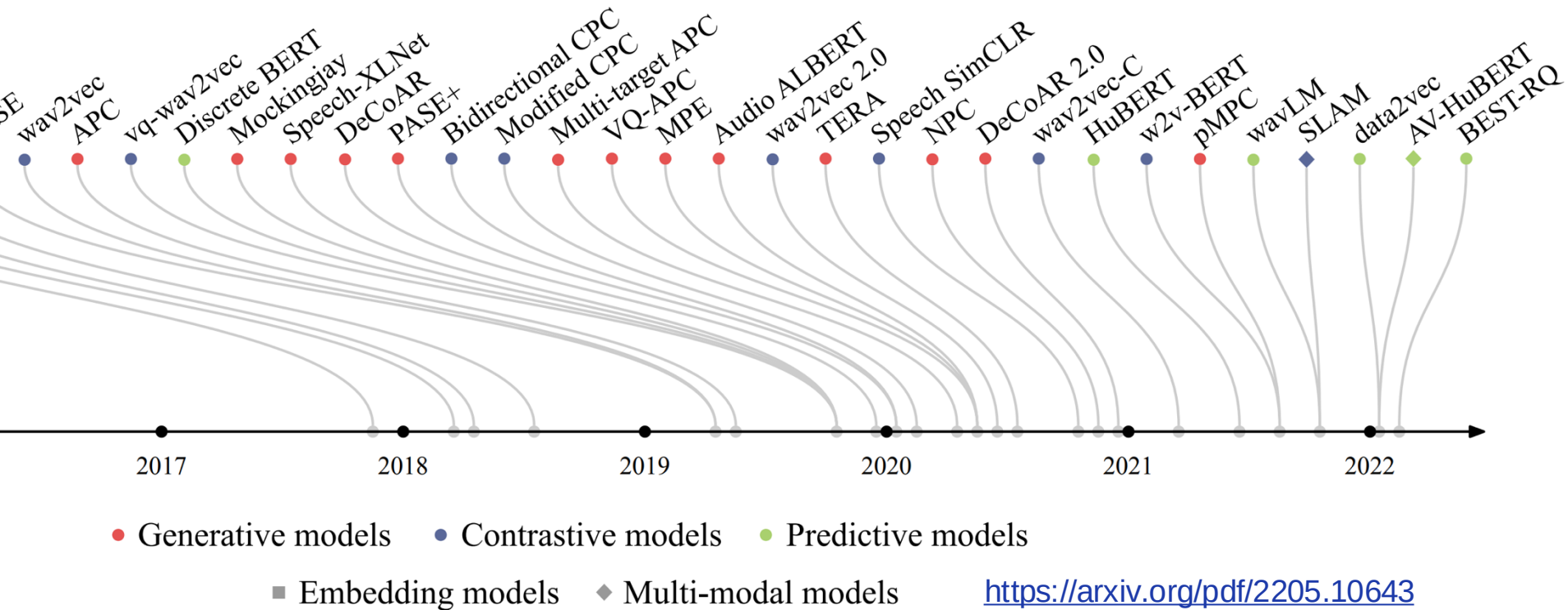


## Phase 2: Downstream



# SSL Model Categorization

A. Mohamed et al. (2022), "Self-Supervised Speech Representation Learning: A Review," in *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1179-1210, Oct. 2022, doi: 10.1109/JSTSP.2022.3207050.



# SSL Model Categorization

| Model Type  | Goal / Objective                                         | Typical Use Cases                          | Examples / Models                               |
|-------------|----------------------------------------------------------|--------------------------------------------|-------------------------------------------------|
| Generative  | Learn to <b>generate</b> raw or structured data          | TTS, ASR decoding, data synthesis          | GPT, Whisper, Tacotron, VALL-E, StableDiffusion |
| Contrastive | Learn to <b>differentiate</b> positive vs negative pairs | SSL pretraining, speaker ID, retrieval     | Wav2vec 2.0 pretrain, SimCLR, CPC               |
| Predictive  | Learn to <b>predict masked/missing parts</b>             | SSL, language modeling, audio modeling     | HuBERT, WavLM, BERT, GPT, data2vec              |
| Embedding   | Learn to <b>encode input into dense vector</b>           | Speaker embedding, clustering, retrieval   | x-vector, ECAPA-TDNN, word2vec, BERT embeddings |
| Multimodal  | Learn to <b>integrate multiple modalities</b>            | ASR+translation, speech-text-image systems | Whisper, Flamingo, CLIP, AudioCLIP, SpeechT5    |

# SSL Model Categorization

| Model Type  | Goal / Objective                                         | Key Technique                                 | Typical Use Cases                          |
|-------------|----------------------------------------------------------|-----------------------------------------------|--------------------------------------------|
| Generative  | Learn to <b>generate</b> raw or structured data          | Reconstruction, auto-regression, diffusion    | TTS, ASR decoding, data synthesis          |
| Contrastive | Learn to <b>differentiate</b> positive vs negative pairs | Contrastive loss (e.g., InfoNCE, SimCLR)      | SSL pretraining, speaker ID, retrieval     |
| Predictive  | Learn to <b>predict masked/missing parts</b>             | Masked modeling, next-step prediction         | SSL, language modeling, audio modeling     |
| Embedding   | Learn to <b>encode input into dense vector</b>           | Projection head, pooling, triplet/cosine loss | Speaker embedding, clustering, retrieval   |
| Multimodal  | Learn to <b>integrate multiple modalities</b>            | Fusion (early/late), cross-attention          | ASR+translation, speech-text-image systems |

# Popular SSL + Transformer Models

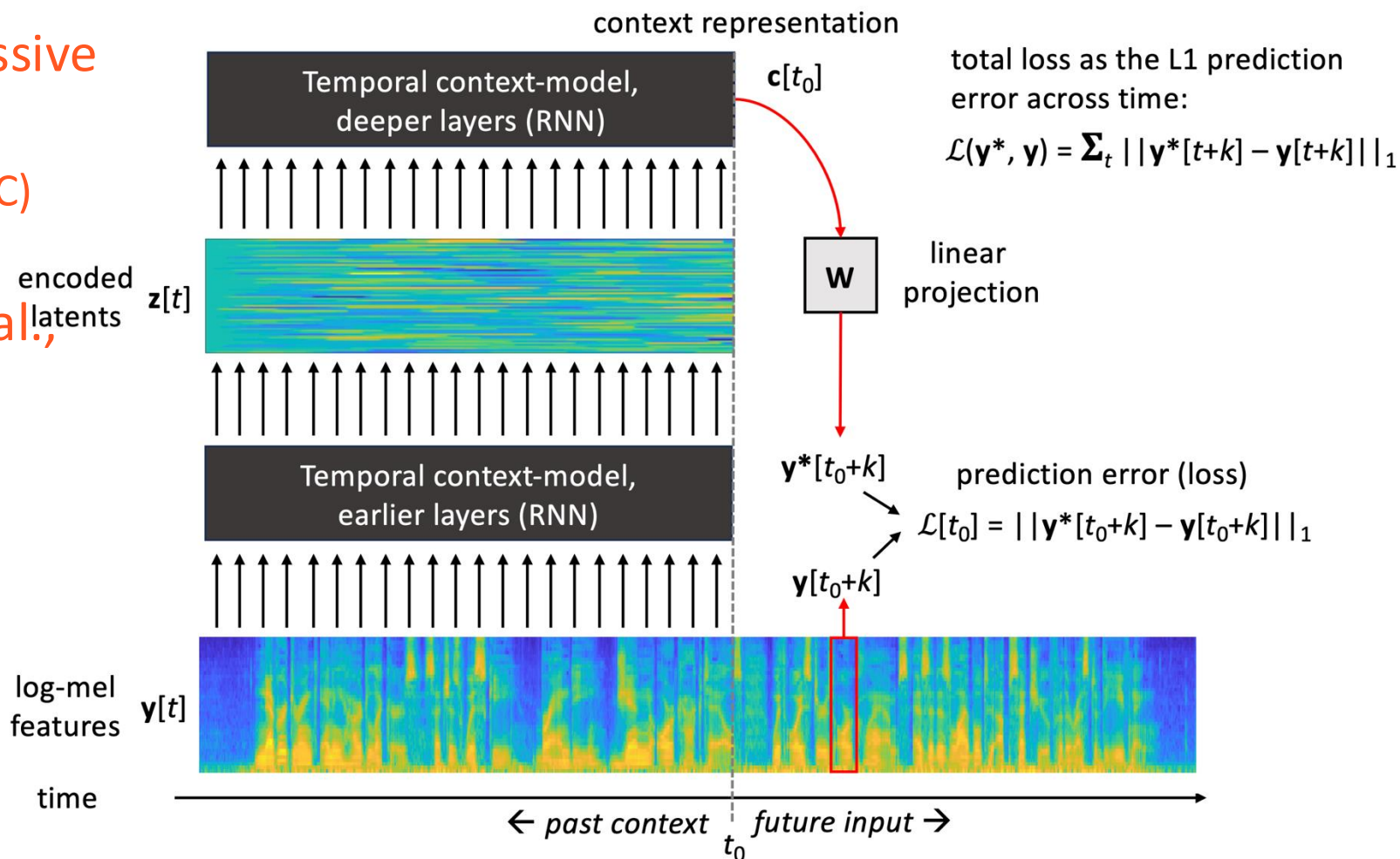
| Model       | Developer          | Highlights                                                    |
|-------------|--------------------|---------------------------------------------------------------|
| Wav2vec 2.0 | Meta (Facebook AI) | SSL from raw audio + Transformer encoder, fine-tuned with CTC |
| HuBERT      | Meta               | Adds hidden unit clustering to improve SSL learning           |
| WavLM       | Microsoft          | Advanced version of HuBERT, better for noisy/realistic speech |
| Whisper     | OpenAI             | Supervised Transformer encoder-decoder, multilingual ASR      |
| Data2vec    | Meta               | Unified SSL model for speech, vision, and text                |

# Popular SSL + Transformer Models

| Model                            | Multilingual?                     | Details                                                                                                                                      |
|----------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Wav2vec 2.0                      | ✗ (originally monolingual)        | Trained on English (LibriSpeech), but can be <b>fine-tuned</b> on other languages                                                            |
| HuBERT                           | ✗ (monolingual)                   | Pretrained mostly on English data; can be adapted to other languages                                                                         |
| WavLM                            | ✗ (mostly English)                | Trained on large English corpora, but supports <b>robust generalization</b> to other languages when fine-tuned                               |
| Whisper                          | ✓ Yes                             | Trained <b>multilingually</b> on 680k hours of speech across <b>96+ languages</b> — supports multilingual ASR and translation out of the box |
| Data2vec                         | ✗ (speech version is monolingual) | Cross-modality SSL, but multilingual support not default for speech                                                                          |
| XLSR (Cross-lingual Wav2vec 2.0) | ✓ Yes                             | Special version of Wav2vec 2.0 trained on <b>53+ languages</b> (CommonVoice, BABEL, etc.)                                                    |

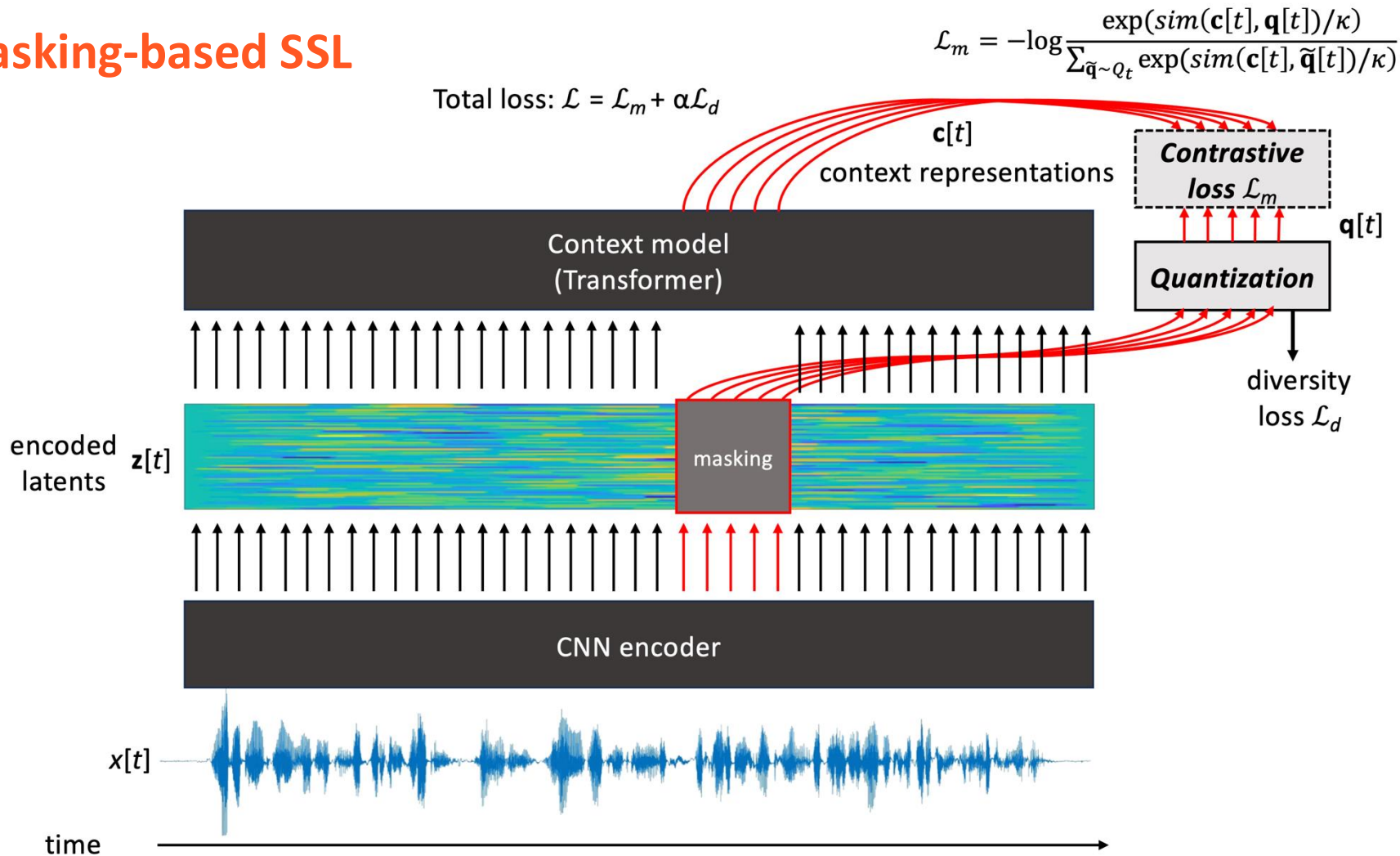


Autoregressive  
Predictive  
Coding (APC)  
algorithm  
(Chung et al.,  
2019)

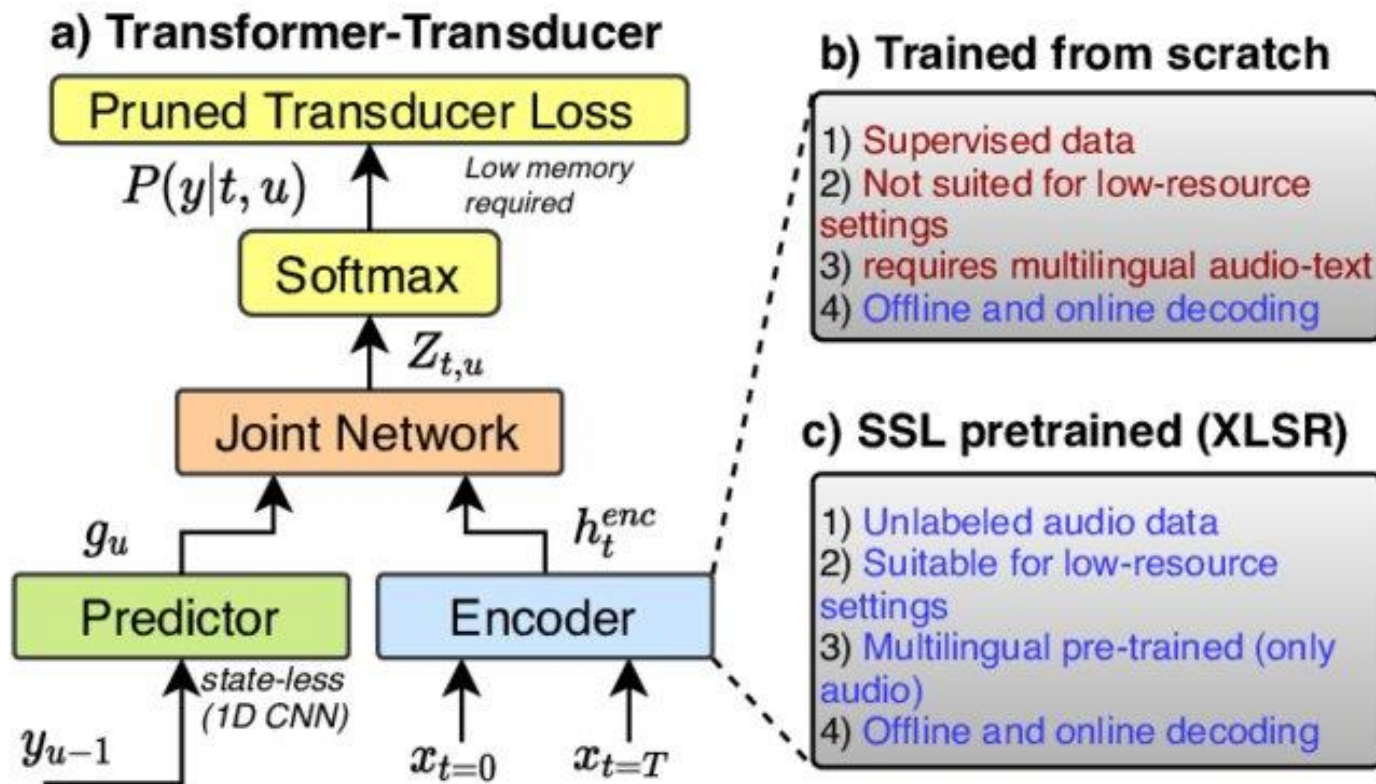


*Predict spectral envelope features (e.g., log-Mel spectra or MFCCs) approximately 50 ms in the future, given access to the current and past observations of the input features*

# Masking-based SSL



# XLSR-Transducer: Streaming ASR for Self-Supervised Pretrained Models (2024)



## 7.4. ASR Challenges & Solutions

- **Accuracy of Recognition:** ASR still struggles with accurately recognizing words or phrases, especially in complex contexts or unclear speech.
- **Language Complexity:** Natural language is diverse and nuanced, posing challenges for ASR in handling various language variations and accents.
- **Context Understanding:** ASR needs improvement in understanding context and deeper meanings of sentences, particularly in complex situations.

## 7.4. ASR Challenges & Solutions

- **Deep Learning Models:** Utilizing deep learning models such as Transformers and their variants has significantly improved ASR accuracy by learning from large datasets and self-adjusting.
- **Optimized Training Data:** Using representative and diverse datasets from multiple sources enhances ASR's recognition capabilities and confidence.
- **Integration with Other Technologies:** Combining ASR with Machine Learning and Natural Language Processing (NLP) helps in comprehensive language understanding and processing.
- **Context-awareness Integration:** To improve context understanding, ASR can integrate methods and models to recognize relationships between words and sentences in text or conversations.

## Specific challenges on...

- **Limited Knowledge of Rare Words:** ASR often struggles with recognizing rare words, specialized terms, or uncommon vocabulary, leading to errors in recognition or misinterpretation of content.
- **Recognition of Proper Names:** ASR faces difficulties in accurately recognizing and pronouncing proper names such as personal names, place names, or company names, especially when these names are uncommon or have different phonetic variations.

## Specific solutions...

- **Expand Training Data:** Incorporate diverse and extensive datasets that include rare words and proper names. This helps ASR learn and improve its ability to recognize these less common terms.
- **Utilize Transfer Learning Techniques:** Apply transfer learning techniques to train ASR on rich and diverse datasets, enabling it to learn how to recognize rare words and proper names effectively.
- **Integrate with NLP Technologies:** Use Natural Language Processing (NLP) methods such as Named Entity Recognition (NER) to enhance the recognition of proper names and specialized terms in various contexts.
- **Enhance Models and Algorithms:** Develop and optimize models and algorithms to ensure more accurate recognition of rare words and proper names, considering their linguistic complexities and variations.

# ASR Trending

- **Self-supervised Learning:** Methods where ASR learns from audio data without detailed labels.
- **Multi-language ASR:** Models that handle multiple languages effectively.
- **Robustness and Adaptability:** Techniques for ASR to work well in diverse environmental conditions.
- **Context-aware ASR:** Models using context to improve accuracy.
- **Personalized ASR:** Customizing ASR for individual communication styles.
- **Integration with AI and NLP:** Combining ASR with AI and NLP to enhance understanding and responsiveness.



# Multilingual datasets for ASR

| Sr No. | Name                                                                                                                                                  | License Type    | Languages     | Hours/Sentence pairs (Approx) |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------|-------------------------------|
|        | <a href="https://www.sciencedirect.com/science/article/pii/S2666307424000573">https://www.sciencedirect.com/science/article/pii/S2666307424000573</a> |                 |               |                               |
| 1      | MuST-C                                                                                                                                                | CC BY-NC-ND 4.0 | 15 Languages  | 385-504 h                     |
| 2      | Europarl                                                                                                                                              | Unknown         | 21 Languages  | 24M sentences                 |
| 3      | MediaSpeech                                                                                                                                           | Open Source     | 4 Languages   | 10 h                          |
| 4      | CoVoST                                                                                                                                                | CC0             | 16 Languages  | 2880 h                        |
| 5      | IWSLT TED                                                                                                                                             | CC BY-NC-ND 4.0 | 109 Languages | Variable Dataset Size         |
| 6      | VoxPopuli                                                                                                                                             | CC0             | 23 Languages  | 400K h                        |
| 7      | Common Voice <sup>1</sup>                                                                                                                             | CC BY-SA 3.0    | 112 Languages | 18.6K h                       |

# Multilingual datasets for ASR

| <b>Sr</b> | <b>Name</b>      | <b>License Type</b> | <b>Languages</b>           | <b>Hours/Sentence pairs</b> |
|-----------|------------------|---------------------|----------------------------|-----------------------------|
| <b>8</b>  | LibriVoxDeEn     | CC BY-NC-SA 4.0     | De-En                      | 100 h                       |
| <b>9</b>  | Microsoft Speech | C-UDA 1.0           | Gujarati, Telugu,<br>Tamil | Insufficient information    |
| <b>10</b> | IIT Bombay       | CC BY-NC 4.0        | En-Hi                      | 1.66M segments              |
| <b>11</b> | MultiIndicMT     | Open Source         | 11 Languages               | 11M sentence pair           |
| <b>12</b> | IndicCorp        | CC BY-NC 4.0        | 12 Languages               | 9B tokens                   |
| <b>13</b> | Dhwani           | MIT License         | 40 Languages               | 17k h                       |
| <b>14</b> | MUCS 2021        | Various licenses    | 6 Languages                | 145 h                       |

## Multilingual datasets for ASR

| <b>Sr No.</b> | <b>Name</b>              | <b>License Type</b>        | <b>Languages</b>   | <b>Hours/Sentence pairs (Approx)</b> |
|---------------|--------------------------|----------------------------|--------------------|--------------------------------------|
| <b>15</b>     | Shrutilipi               | CC0                        | 12 Languages       | 6457 h                               |
| <b>16</b>     | Vistaar                  | MIT License                | 12 Languages       | 10736 h                              |
| <b>17</b>     | NITK-IISc                | Open Source                | 6 Languages        | Insufficient information             |
| <b>18</b>     | Voxforge                 | GNU General Public License | 6 Languages        | Insufficient information             |
| <b>19</b>     | Tatoeba                  | Various CC licenses        | Over 100 Languages | Variable Dataset Size                |
| <b>20</b>     | FLEURS                   | CC BY 4.0                  | 102 Languages      | 1.4k h                               |
| <b>21</b>     | Multilingual LibriSpeech | CC BY 4.0                  | 8 Languages        | 44.5k h (Eng), 6k h (other)          |

Q/A



<https://app.sli.do/event/wPmQxU18pEZmnxHpv1Uo3m>

# Final Test



<https://forms.office.com/r/UAbcquMvTa?origin=|prLink>